Paleomagnetism of Late Archean metavolcanics and metasediments, Abitibi orogen, Canada: tholeiites of the Kinojevis Group

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Iron-rich and magnesium-rich basaltic tholeiites of the latest Archean Kinojevis Group, central Abitibi Belt, northeastern Ontario, contain a natural remanent magnetization that is generally multivectorial. The components in all units reside in essentially pure magnetite, not the original titanomagnetites (where x for $Fe_{3-x}Ti_xO_4$ probably was 0.60–0.65). Any original TRM was lost by chemical reconstitution of the magnetic oxides in response to primary deuteric conditions, long-term burial, regional metamorphism to prehnite–pumpellyite facies, and possibly intrusion by Matachewan dikes. Data from contact tests with Matachewan dikes indicate that the units are indeed capable of retaining a very latest Archean – earliest Proterozoic field (e.g., $D = 194.9°$, $I = -14.3°$, $k = 8.1$, $\alpha_{95} = 7.9°$; $n = 45$ vectors, 38 samples). High-coercivity, high-blocking-temperature directions from samples from other flows, corrected for nearly penecontemporaneous downwarping, are in only crude agreement with those of Matachewan dikes, possibly suggesting that these components reflect a general Late Archean – Early Proterozoic field for the Superior Province. The paleomagnetic data from Kinojevis tholeiites indicate the emplacement, burial, and tight downwarping of the 10 km or so of Kinojevis stratigraphy were nearly synchronous with Matachewan intrusion. The Kinojevis data by themselves cannot be taken as statistically reliable indicators of the pre- (or immediately post-) Matachewan geomagnetic field nor can temperature–time relations for magnetic viscosity be used to predict the preservation of a statistically reliable TRM in any of these units. Individual magnetization components were blocked over geologically short periods of time, whereas the ensemble of data from discrete flows must record secular variations, field excursions, and possibly long-term polar wander.

Les tholéiites basaltiques riches en fer et riches en Mg du groupe Kinojevis de la fin de l'Archéen, dans la ceinture centrale de l'Abitibi, au nord-est de l'Ontario, renferment une aimantation rémanente naturelle généralement multi-vectorielle. Dans toutes les unités les composantes sont portées par une magnétite essentiellement pure et non par la titanomagnétite originale (où la valeur de x dans $Fe_{3-x}Ti_xO_4$ était probablement de 0,60 à 0,65). Toute aimantation rémanente thermique fut perdue lors de la reconstitution chimique des oxydes magnétiques par réaction aux conditions deutériques primaires, à une longue période d'enfouissement et au métamorphisme régional du faciès préhnite–pumpellyite et possiblement à l'intrusion des dykes Matachewan. Les données provenant des essais pour les contacts des dykes Matachewan indiquent que ces unités possédaient réellement le pouvoir de retenir le champ de la fin de l'Archéen – début du Protérozoïque (exemple: $D = 194{,}9°$, $I = -14{,}3°$, $k = 8{,}1$, $\alpha_{95} = 7{,}9°$; $n = 45$ vecteurs, 38 échantillons). Une force de coercition forte, une température de blocage élevée des directions d'échantillons provenant d'autres coulées, même après correction pour la subsidence presque pénécontemporaine, ne constituent que des preuves grossières qui concordent avec celles des dykes Matachewan, suggérant possiblement que ces composantes reflètent un champ global d'âge Archéen supérieur – Protérozoïque inférieur dans la province du Supérieur. Les données paléomagnétiques des tholéiites Kinojevis restreignent la mise en place, l'enfouissement et la subsidence serrée de plus ou moins 10 km d'épaisseur stratigraphique Kinojevis à s'être produits presque en même temps que l'intrusion Matachewan. Les données sur le groupe Kinojevis en elles-mêmes ne doivent pas être considérées comme des indices statistiquement fiables du champ géomagnétique qui existait avant (ou immédiatement après) l'intrusion Matachewan pas plus que les relations température–temps pour la viscosité magnétique peuvent servir à prédire statistiquement la mémorisation de l'aimantation rémanente thermique dans n'importe quelle de ces unités. Les composantes individuelles d'aimantation furent bloquées durant de courtes périodes de temps géologique, tandis que l'ensemble des données des coulées séparées doit avoir enregistré les variations séculaires, les déviations de la courbe du champ et possiblement la dérive des pôles à long terme.

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Introduction

Investigations of the Archean geomagnetic field for North America have concentrated on intrusive igneous bodies, e.g., dikes (Irving and Naldrett 1977; Ernst 1981; Schutts 1980; Fahrig *et al.* 1965; Strangway 1964; Schutts and Dunlop 1981), stocks and batholiths (Dunlop and Buchan 1975, 1976; Dunlop 1979; Irving and Naldrett 1977), and sills and laccoliths (Bergh 1970; Ueno and Irving 1976; Irving and Naldrett 1977; Geissman *et al.* 1982). The units are relatively well dated, and either have not been deformed subsequent to

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emplacement or have relatively simple tectonic histories, although it is difficult to prove this for batholiths (e.g., Beck 1980). In addition, properties of the natural remanent magnetization (NRM) of these lithologies suggest the presence of a (primary) thermoremanent magnetization (TRM). Compilation of apparent polar wander (APW) paths from these data for the Archean (e.g., Irving 1979; Schutts and Dunlop 1981) is therefore straightforward.

Success in compilation of APW paths requires that the field be recorded as continuously as possible by the rock record. The intermittent nature of Archean plutonism, however, makes this task difficult. Alternatively, volcanic units fill many gaps in the Archean rock record (Goodwin and Ridler 1970; Goodwin 1974, 1978, 1980). Although extrusives constitute a significant portion of Archean terranes, their paleomagnetism has been studied only infrequently and those few studies have often been unsuccessful in deciphering any totally consistent record (Halls 1975; Pesonen 1973; Schwarz 1966, 1977; D. J. Dunlop, personal communication, 1980), though exceptions exist (e.g., Ridler and Foster 1981). The TRM in Archean metavolcanics, if it is retained and can be isolated (e.g., McElhinny and Senenayake 1980), might be used to fill in important gaps in Archean APW paths. "Primary" and secondary magnetizations in Archean metavolcanics may be of benefit in deciphering complex structures and in exploration for massive sulfide deposits by explaining the effects of ocean floor alteration, regional metamorphism, and massive sulfide mineralization on the NRM.

In this and in companion contributions (Geissman *et al.* 1982, and in preparation; Letros *et al.* 1983; Tasillo-Hirt *et al.* 1982) we describe the NRM, magnetic properties, and aeromagnetic signature of metavolcanics and metasediments in the central Abitibi Greenstone Belt. Between Kirkland Lake and Lake Abitibi (Fig. 1), the low degree of deuteric alteration and dynamothermal metamorphism (Dimroth and Lichtblau 1979) and the near absence of later intrusive activity all, in theory, imply the retention of at least the high-blocking-temperature portion of a TRM. The relatively simple tectonism of some units may allow corrections to be made for structure and "primary" magnetizations to thus be included in APW paths. Schutts and Dunlop (1981) reported magnetization data from "basalts and gabbros" in the central Abitibi Belt that they suggested are representative of a Late Archean geomagnetic field existing during volcanism and plutonism. They did not correct their data for regional structure, citing arguments that "structure in the volcanics developed during deposition" and quoting paleomagnetic data from sill-like gabbros yielding negative fold tests. This paper is concerned with tholeiitic basalts of the Kinojevis Group (Jensen 1978*a*, 1979, 1981). We studied the paleomagnetism of four geologically and petrologically well characterized "sites" in Kinojevis flows in order to determine in detail the nature of their NRM. Detailed mapping indicates that flows at all of these areas were deposited horizontally and deformed only by simple tilting. A "primary" remanence, if indeed one exists, should be corrected for regional structure.

General geology

The area sampled (Fig. 1) is part of Goodwin and Ridler's (1970) Noranda–Benoit volcanic complex, in the central Abitibi Belt. As this paper represents our initial studies of the paleomagnetism of this belt, the geology of the area is described in relative detail here; in other papers only the most pertinent aspects are included. The Ontario portion of the complex is composed of three subalkalic volcanic cycles, each associated with sedimentation and intrusion of comagmatic units (Jensen 1978*a*, 1981). The stratigraphic sequence is illustrated in Fig. 2. The Kinojevis Group, of the second volcanic cycle, is well preserved on both limbs of a major east–west-trending synclinorium in the Noranda–Benoit volcanic complex. At the base of the second cycle, flows of ultramafic and basaltic komatiite composition are interlayered with calc-alkalic tuff and tuff breccias, conglomerates, greywackes, argillites, cherts, and iron formation. Ultramafic and mafic members of this sequence, collectively termed the Stoughton–Roquemaure Group in the north and the Larder Lake Group in the south (Jensen 1979), become predominant upwards in the section. The Stoughton–Roquemaure Group is approximately 10 km in thickness and is conformably overlain by the Kinojevis Group. The Kinojevis Group consists of a 10 km thick succession of alternating magnesium- and iron-rich tholeiitic basalts, with minor andesites, dacites, and rhyolites. Flows average about 65 m in thickness and are often traceable for several kilometres. They generally consist of a basal zone of flattened glass shards (hyaloclastite) and spherulites, a central zone of coarse cumulate material, overlying granular basalt, and an often highly vesiculated, shard-rich flow top (Pearce and Birckett 1974; Jensen 1978*b*; Jackson 1980*a*). Individual flows either alternate between iron- and magnesium-rich compositions or are grouped into sequences of several iron- or magnesium-rich units. Pearce and Birckett (1974) suggested on petrologic and geochemical groups that each flow has experienced some degree of *in situ* differentiation.

Above the Kinojevis lavas is 12 km of calc-alkaline volcanics and hypabyssal intrusives, which occupy the center of the synclinorium. These constitute the Blake River Group; they were formed by the partial melting of ultramafic and mafic rocks of the lower komatiitic and tholeiitic volcanics (Jensen 1981).

The third volcanic cycle is preserved on both the south

limb of the synclinorium, along the Kirkland Lake – Larder Lake Fault Zone, and on the north limb, along the Destor–Porcupine Fault Zone (Fig. 1). In the south, approximately 3 km of fluvial sediments and turbidites, trachytes, and syenite stocks (collectively the Timiskaming Group) is either in fault contact or unconformably overlies volcanics of the second cycle.

North–south-trending Matachewan diabase dikes cut all volcanic cycles. However, the dikes, usually recognized by their aeromagnetic anomalies, are rare to non-existent in the central part of the Noranda–Benoit synclinorium (Lumbers and Milne 1978; Pyke *et al.* 1971). Their concentration increases westward from a point approximately 10 km east of Highway 11, reaching a maximum density (approximately two dikes per kilometre) west of Timmins.

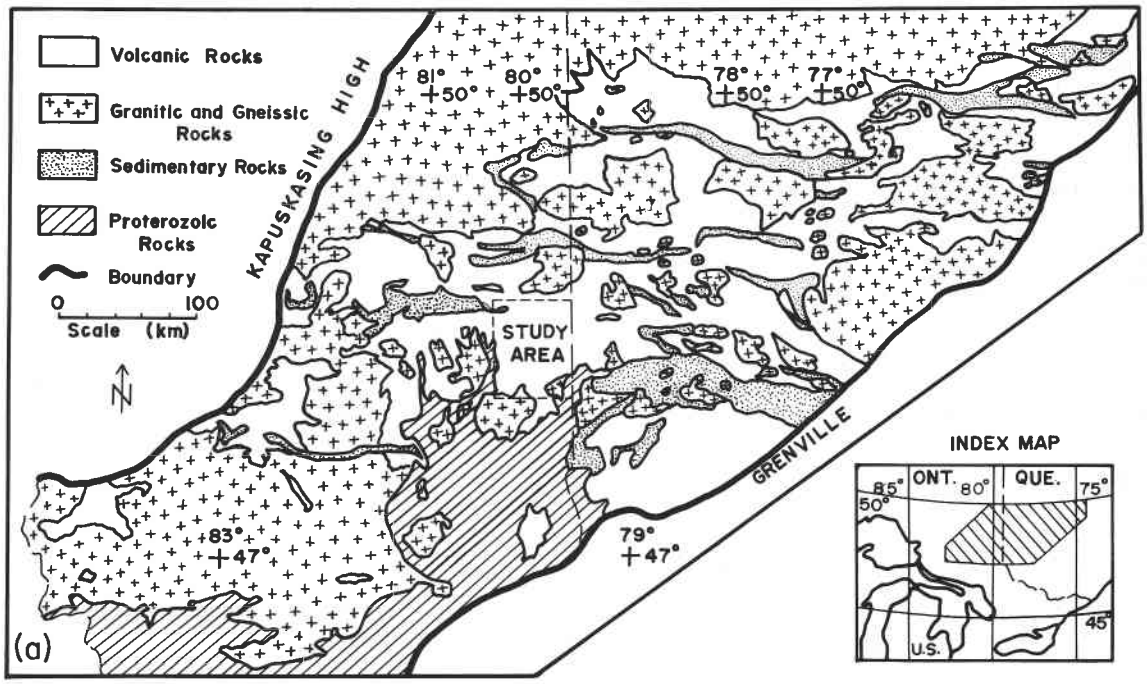
Letros *et al.* (1983) have described the aeromagnetism of the area between Lake Abitibi and Kirkland Lake. The Kinojevis Group, in particular, appears as a conspicuous set of alternating high and low east-northeast-trending magnetic anomalies, corresponding to iron-rich and magnesium-rich flow sequences, respectively.

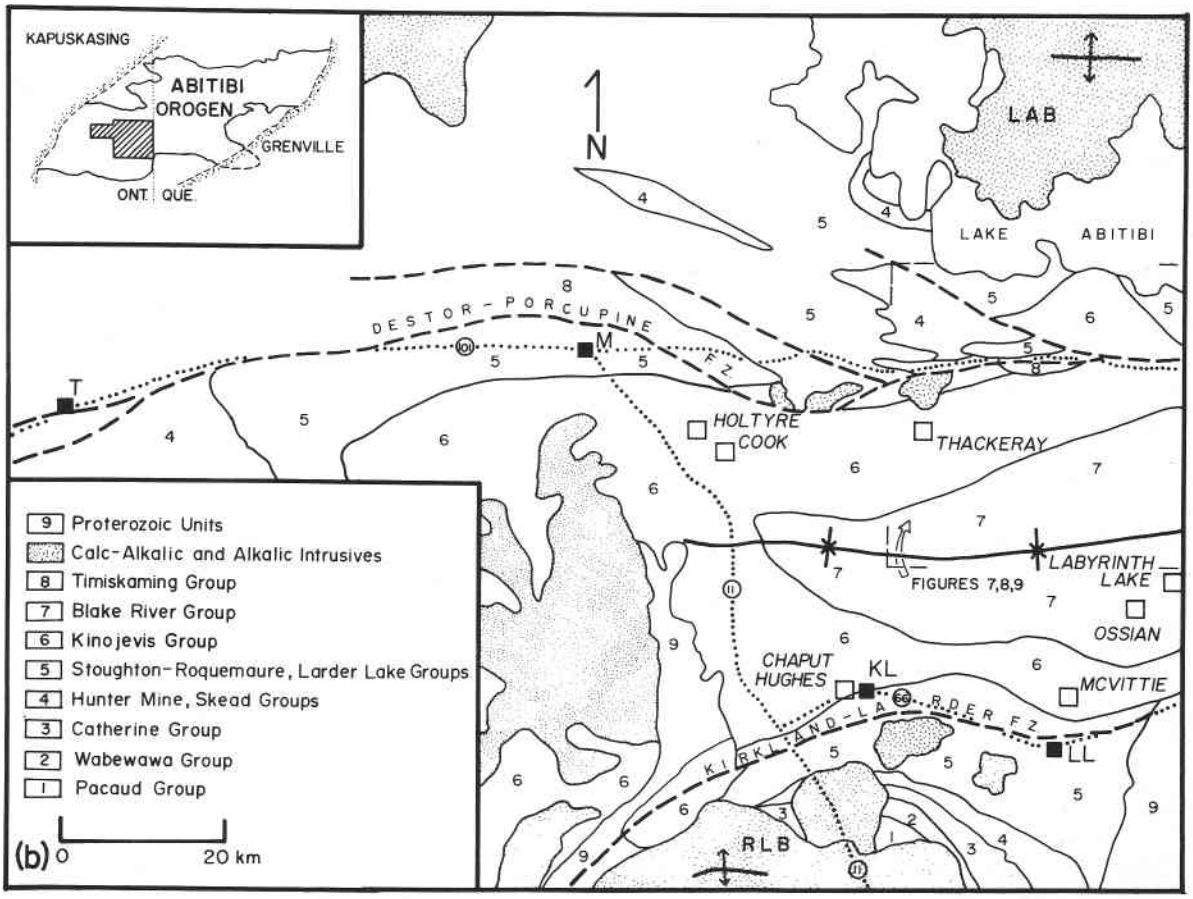
The depositional and structural history of the Noranda–Benoit volcanic complex has been interpreted in several ways (Goodwin and Ridler 1970; Jensen 1978*a*, 1979, 1981; Goodwin 1980). Flows of the first and second volcanic cycles appear to have been extruded within a shallow, ocean basin environment (Jensen 1981). Flow thicknesses and percentages of silicic flows increase toward the center of the synclinorium on both limbs, suggesting that the source area was near the fold axis. Kinojevis flows were deposited on mild topography and were tilted downward in the center of the synclinorium to accommodate additional lavas. With Blake River magmatism, an irregular topography was created by volcanic centers near the fold axis. Accumulation of approximately 30 km of lavas and hypabyssal intrusives and the withdrawal of a sizeable magma chamber at depth ultimately caused the gradual downwarping of the volcanic complex into its present form. A few kilometres north and south of the syncline axis, orientations of flows as measured on compressed shards and pillows are nearly vertical. Contacts between Kinojevis flows are generally conformable and thus major downwarping of the complex was continuous until the waning stages of Blake River magmatism. Vertical movements of several hundreds of metres occurred along the Kirkland Lake – Larder Lake and Destor–Porcupine Fault Zones. These faults were probably major mechanisms for subsidence during the deposition of the second volcanic cycle (Jensen 1981). During Timiskaming deposition, both fault zones acted as graben structures, trapping steeply dipping units within a narrow belt. Emplacement of Matachewan dikes followed development of the synclinorium.

There is evidence to suggest that at least some of the units in some areas of the Noranda–Benoit volcanic complex may have preserved a detectable TRM. In addition, there are sufficient age data from these units and the field relations are such that proof (or disproof) of the existence of a TRM is achievable. The prehnite–pumpellyite to subgreenschist facies (Jolly 1974, 1980) of regional and (or) burial metamorphism of the terrane led Irving and Naldrett (1977) to suggest that a TRM might be preserved if much of it were contained in high-blocking-temperature phases. Metamorphism, in more detail, is complex in history, and at least six phases have been distinguished in units of the central and eastern parts of the complex (Dimroth *et al.* 1974). Pearce and Birckett (1974) indicated that temperatures never exceeded 400°C in any metamorphic event and were probably much less. Volcanic textures, major portions of the original magmatic mineralogies, and original within-flow chemical variations are still preserved, especially in Kinojevis flows (Pearce and Birckett 1974; Jensen 1978*a*; Dimroth and Lichtblau 1979). In addition, in some parts of Kinojevis flows the original magnetic mineralogy is still preserved.

Units of the second and third volcanic cycles were emplaced within a well bracketed, short time period. Dacites underlying Stoughton–Roquemaure units east of Munro Township (Fig. 1) give a U–Pb zircon date of 2710 ± 2 Ma, and volcanic and subvolcanic units forming the top of the Blake River pile give an age of 2703 ± 2 Ma (Nunes and Jensen 1980). Zindler *et al.* (1978) listed a Sm–Nd age of 2765 ± 47 Ma for komatiites and related intrusives of the Stoughton–Roquemaure Group, Munro Township. The Rb–Sr age of Matachewan dikes is 2690 ± 93 Ma (Gates and Hurley 1973), in overall agreement with field relations, though most workers infer that the dikes are closer to 2600 Ma in age (e.g., Irving 1979; Schutts and Dunlop 1981). A small difference in age, if real, between the volcanic cycles of the Noranda–Benoit volcanic complex and Matachewan activity implies a similarity or near similarity in TRM (or depositional remanent magnetization (DRM)) directions among these various units. Post-depositional tectonism of the volcanics and sediments, prior to Matachewan intrusion, should have dispersed the directions. In turn, the preservation of a TRM in the volcanics and sediments might be implied if,

FIG. 1. Generalized geologic map of the central part of the Abitibi Greenstone Belt. (*a*) Regional. (*b*) More local. Modified from Jensen (1979). LAB = Lake Abitibi batholith; RLB = Round Lake batholith; KL = Kirkland Lake; LL = Larder Lake; T = Timmins; M = Matheson.


Volcanic Rocks
Granitic and Gneissic Rocks
Sedimentary Rocks
Proterozoic Rocks
Boundary
0 100
Scale (km)
KAPUSKASING HIGH
81° +50°
80° +50°
78° +50°
77° +50°
STUDY AREA
GRENVILLE
83° +47°
79° +47°
(a)
INDEX MAP
85° ONT. 80° QUE. 75°
50°
45°
U.S.


KAPUSKASING
ABITIBI OROGEN
GRENVILLE
ONT. QUE.
N
LAB
LAKE ABITIBI
DESTOR-PORCUPINE FZ
T
M
HOLTYRE
COOK
THACKERAY
FIGURES 7,8,9
LABYRINTH LAKE
OSSIAN
CHAPUT
HUGHES
KL
MCVITTIE
KIRKLAND-LAKE-ROUYN FZ
LL
RLB
9 Proterozoic Units
Calc-Alkalic and Alkalic Intrusives
8 Timiskaming Group
7 Blake River Group
6 Kinojevis Group
5 Stoughton-Roquemaure, Larder Lake Groups
4 Hunter Mine, Skead Groups
3 Catherine Group
2 Wabewawa Group
1 Pacaud Group
(b) 0 20 km

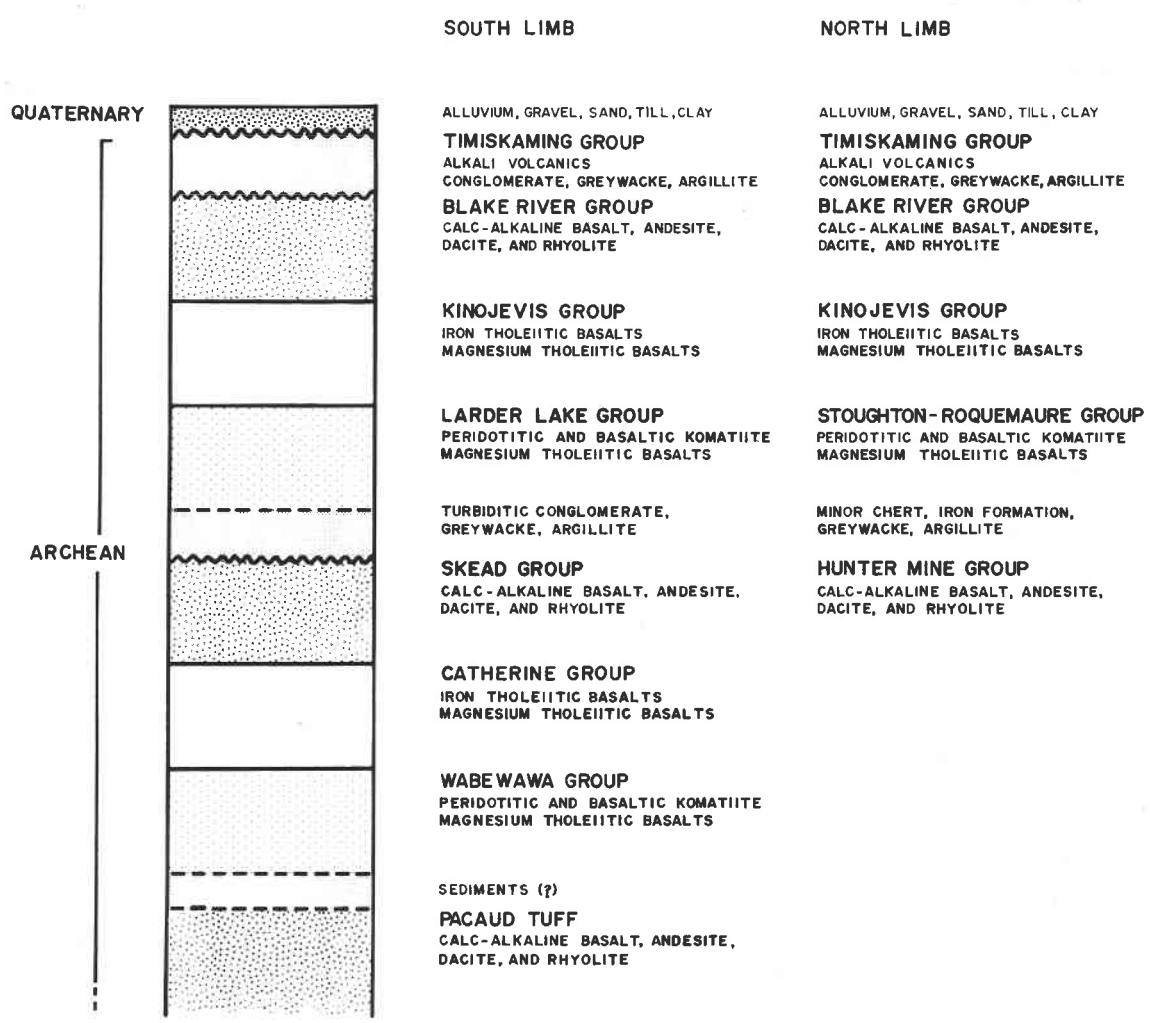


FIG. 2. Geological columns of the north and south limbs of the central Abitibi Belt synclinorium. Modified from Jensen (1979, 1981).

after correction for pre-Matachewan tectonism, the magnetization directions are nearly identical to those from Matachewan dikes.

The structural history of the second and third volcanic cycle units is relatively straightforward. Development of the major east–west-trending synclinorium structure occurred during the deposition of the Kinojevis and Blake River Groups (Jensen 1981). *In situ* "primary" magnetization data from Kinojevis and older units should be corrected for structure because lava successions have been tilted into monoclinal sequences well after emplacement and original cooling.

The data presented here and in companion papers (Tasillo-Hirt *et al.* 1982; Geissman *et al.* 1982, and in preparation) are results of our efforts to extract further information on the Archean field from units of the Abitibi Greenstone Belt. We assumed that an original, syngenetic component of magnetization of statistical significance was present in the lithologies examined, and tests are applied for the presence of this remanence in as many instances as possible.

Methodology

Samples were collected by field drilling techniques or by orienting blocks. A magnetic compass was used for orientation (accuracy of 1–2°). Generally, two to five specimens (2.5 cm long by 2.5 cm diameter) were prepared from each core or block sample. Details of the sampled areas are given with the paleomagnetic data.

The remanent magnetization was generally measured with the prototype SCT rock magnetometer (Goree and Fuller 1976) having a 4 in. (10 cm) diameter vertical access and background noise of around 10^{-7} emu (10^{-10} A m^2). Specimens with greater intensities were measured on a portable Schonstedt magnetometer (model PSM-1) or a Schonstedt spinner magnetometer (model SSM-1a).

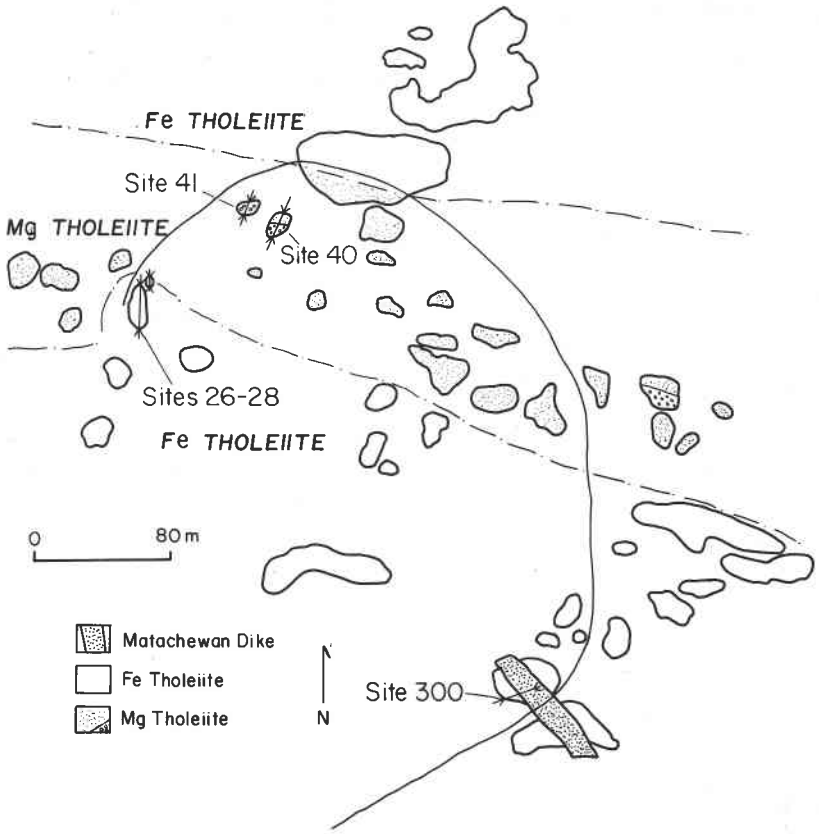


FIG. 3. Generalized geologic map of part of Lava Flow Mountain, Cook Township (Fig. 1). Modified from Jackson (1980*a*).

These were eventually measured on the SCT when their moment was reduced sufficiently in demagnetization. Progressive alternating field (AF) demagnetization was carried out with a single-axis Schonstedt demagnetizer, model GSD-1. A Schonstedt thermal demagnetizer, model TSD-1, was used for most thermal demagnetizations, but two home-built furnaces (one capable of heating in vacuums less than 10^{-3} Torr (0.13 Pa)), were also employed. Hysteresis and Curie temperature properties were measured on a PAR vibrating sample magnetometer. For thermomagnetic runs, whole-rock samples or magnetic separates were run under a vacuum of about 10^{-3} Torr (0.13 Pa) or less. An isothermal remanent magnetization (IRM) was imparted with an electromagnet capable of producing fields up to 1.8 T. Polished thin sections from representative samples of all lithologies were prepared.

All components of the NRM in all samples were identified using progressive demagnetization diagrams (Zijderveld 1967; Dunlop 1979). For the vast majority of samples, magnetization components decayed in a linear fashion in demagnetization, and vector subtraction methods could thus be used to calculate the direction of each component. Because a multivectorial NRM was generally contained in each sample and, as will be shown, the NRM from sample to sample was anything but uniform, demagnetization procedures were generally quite lengthy. At high peak fields in AF demagnetization (generally greater than 50 mT), some samples acquired a spurious component of magnetization, often parallel to the last axis demagnetized. The method of French (1976) was used to verify that this component was usually an anhysteretic remanent magnetization (ARM); this method entails a six-step demagnetization of the specimen at any given field, assuming that the ARM is essentially acquired parallel to the last axis demagnetized. The method was consequently employed where necessary to adequately resolve linear decays of NRM components.

Results

Sampling in the Kinojevis tholeiites is concentrated in three areas selected on the basis of exposure, amount of geologic mapping and petrologic investigations, accessibility, and location with respect to the synclinorium axis and Matachewan dikes.

Cook Township

The first area is Lava Flow Mountain, Cook Township (Fig. 1) where Jackson (1980*a*,*b*) has

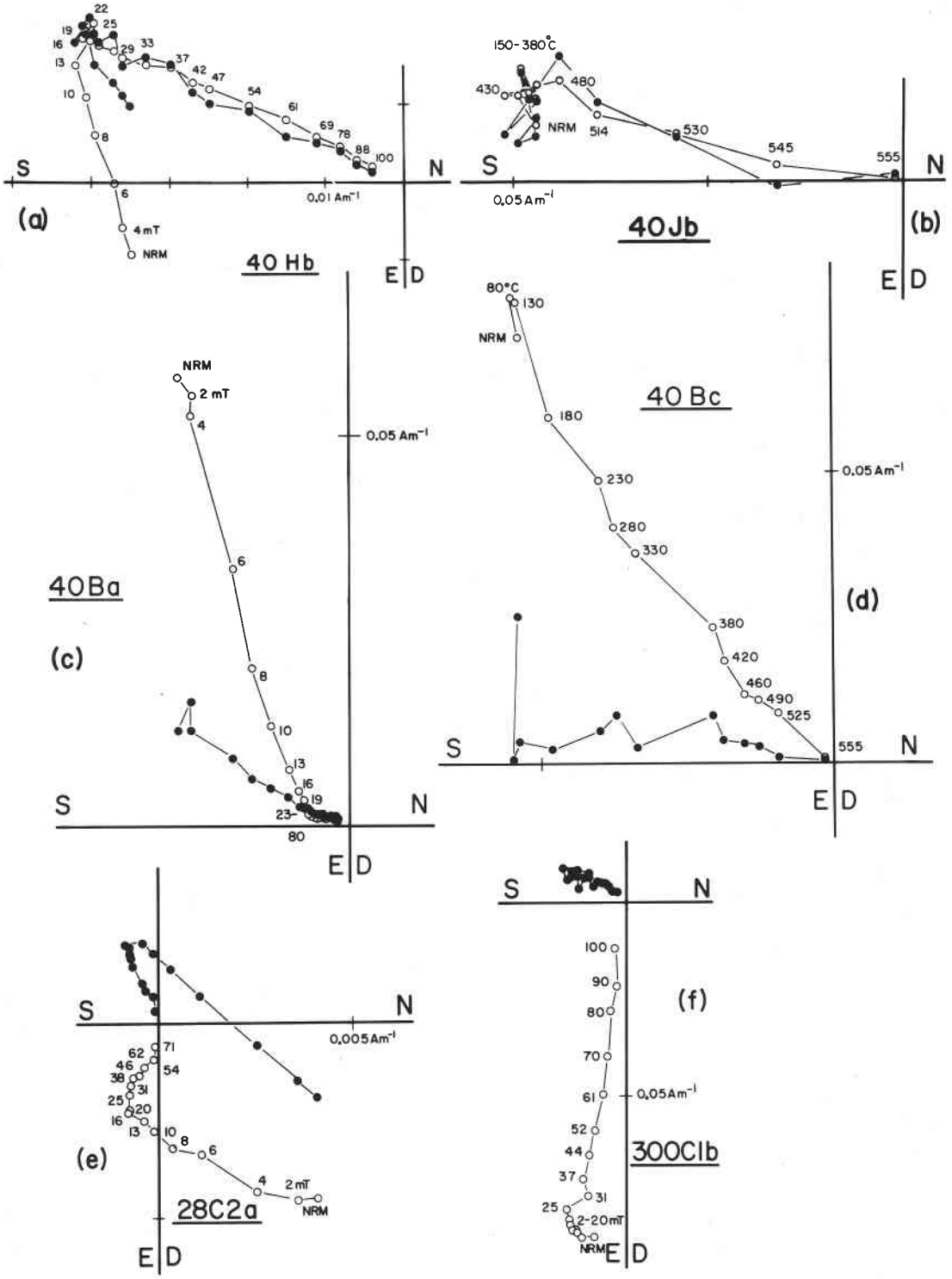


FIG. 4. Progressive orthogonal demagnetization diagrams (Zijderveld 1967) of the successive end point of the magnetization vector during demagnetization, plotted with respect to *in situ* coordinates. Solid (open) circles represent projections on the horizontal north–south (vertical) plane. Peak alternating fields (in mT) or temperatures (in °C) are given beside the vertical projections. Demagnetizations of (*a*, *b*, *c*, *d*) massive, non-porphyritic low-iron tholeiite specimens from samples H, J, and B at site 40; (*e*) basal high-iron tholeiite specimen from sample C at site 28; and (*f*) high-iron tholeiite specimen from sample C at site 300, adjacent to the Matachewan dike.

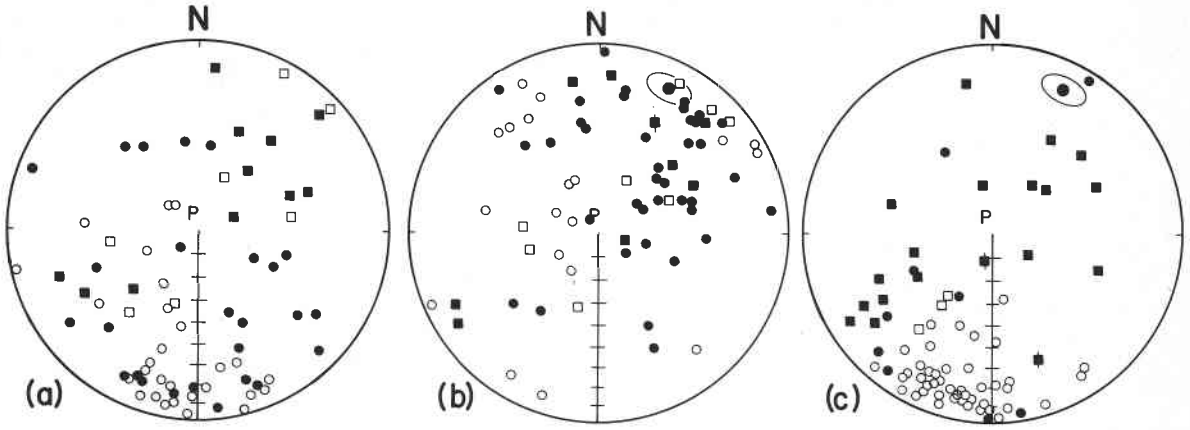


FIG. 5. Paleomagnetic data (*in situ*, equal-area net) from Lava Flow Mountain sites. Open (closed) symbols refer to projections on the upper (lower) hemisphere. (*a*) NRM directions. (*b*) Any secondary directions. (*c*) Ultimate directions. Circles (squares) refer to directions of magnetization components from Kinojevis Mg (Fe) tholeiites. Squares with tick marks indicate directions from samples 300C and 300D, from a Kinojevis Fe tholeiite flow directly adjacent to the Matachewan dike cutting this area. In (*b*) and (*c*), large circles and associate cones of 95% confidence refer to normal polarity Matachewan dike directions (Irving and Naldrett 1977).

mapped a series of alternating Kinojevis flows cut by two Matachewan dikes. Portions of a low-iron flow, with a glomeroporphyritic horizon (41 samples, sites 40 and 41, Fig. 3), and a high-iron flow (20 samples, sites 26–28, Fig. 3) were sampled in detailed traverses. In addition, two samples from a Matachewan dike and six samples from within 10 m of the dike were collected (site 300, Fig. 3).

For the low-iron flow, most samples generally possessed two components of magnetization, revealed by either AF or thermal methods. For example, in Fig. 4*a*, a secondary component of northeast declination and moderate positive inclination is removed by 17.5 mT and thereafter a south-southwest declination, shallow negative-inclination component, trending toward the origin, is removed. In thermal demagnetization, similar results are obtained (Fig. 4*b*). A south-declination, generally shallow negative-inclination component, trending linearly toward the origin, was resolved in the majority of samples from the low-iron flow. In a few samples, however, only AF demagnetization resolved both components (Fig. 4 *c* and *d*). We conclude that the blocking-temperature spectra of the two components in some samples overlap significantly, whereas the coercivity spectra do not.

Samples from the high-iron flow, on the other hand, contain a varied number of magnetization components (Fig. 4 *e* and *f*). Many high-iron samples are characterized by magnetization components of lower median destructive fields than those in the low-iron flow.

At site 300, the two samples of the Matachewan dike have intensities (approximately 0.3 A m^{-1}) typical of those reported elsewhere (Strangway 1964; Fahrig *et al.* 1965; Irving and Naldrett 1977; Ernst 1981; Schutts and Dunlop 1981) and reveal typical south-southwest-declination, shallow negative-inclination components in demagnetization, similar to those at sites 40 and 41.

Magnetization directions from the three groups of sites are shown in Fig. 5. Directions of the components trending toward the origin in the low-iron flow are statistically identical to the reverse polarity TRM directions from Matachewan dikes reported by the workers listed above. The mean of 38 samples (45 vectors) has a value of $D = 194.9°$, $I = -14.3°$, $k = 8.1$, $\alpha_{95} = 7.9°$. Although this is not a contact test in the strictest sense, it can be argued that the well grouped component in most parts of the low-iron flow was acquired during thermal resetting by the Matachewan dike striking toward the outcrop and exposed about 200 m to the south-southeast (Fig. 3). This component is probably not a chemical remanent magnetization (CRM) acquired during Matachewan dike intrusion, as might be expected, because (1) in most cases both the blocking-temperature and coercivity spectra of this component do not overlap the spectra of secondary components and (2) there are no significant differences between the magnetic mineralogy of site 40 and site 41 samples and similar low-iron flow lithologies far removed from Matachewan dikes. The high-blocking-temperature, high-coercivity components and therefore the magnetic mineralogy existing in the low-iron flow at the time of intrusion accurately record the geomagnetic field during the time of Matachewan intrusion. Barring changes in the magnetic mineralogy, we would expect components of equal or greater coercivity and (or) blocking temperature from equivalent flows far removed from Matachewan dikes to reflect an earlier field. This argument assumes, of course, that the high-blocking-temperature, high-coercivity components were acquired

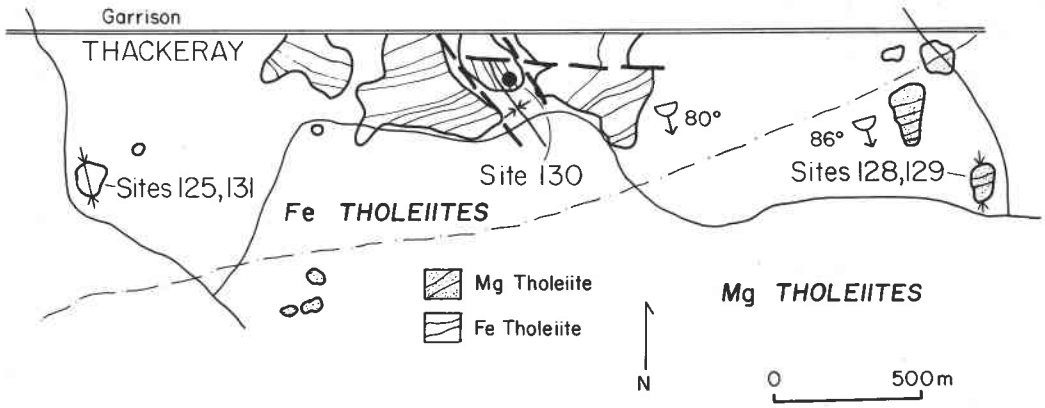


FIG. 6. Generalized geologic map of the northern part of Thackeray Township (Fig. 1). Modified from Jensen (1978*a*).

prior to Matachewan intrusion (e.g., a TRM) and not at a later period of time (e.g., a CRM). Some secondary components isolated at sites 40 and 41 have blocking-temperature spectra that partially overlap with the spectra of the higher blocking-temperature, higher coercivity (Matachewan) direction. We argue that these components are probably CRM's, because two different partial TRM components would not exhibit an overlap of blocking-temperature spectra as described above.

The components trending toward the origin in samples of the high-iron flow, both at sites 26–28 and site 300, are clearly not Matachewan dike directions (Fig. 5). Because their coercivities and blocking temperatures are similar to the well grouped components in the low-iron flow, it is thought that these components may record an earlier field direction and were not reset by the dike. We have no explanation for the negative contact test results (no magnetization component parallel to the Matachewan dike direction) from samples 300C and 300D (Fig. 5*c*), high-iron samples immediately adjacent to the Matachewan dike. The components trending toward the origin in these samples are of similar if not equal coercivities (and presumably blocking temperatures) to those in the low-iron flow (Fig. 4*f*).

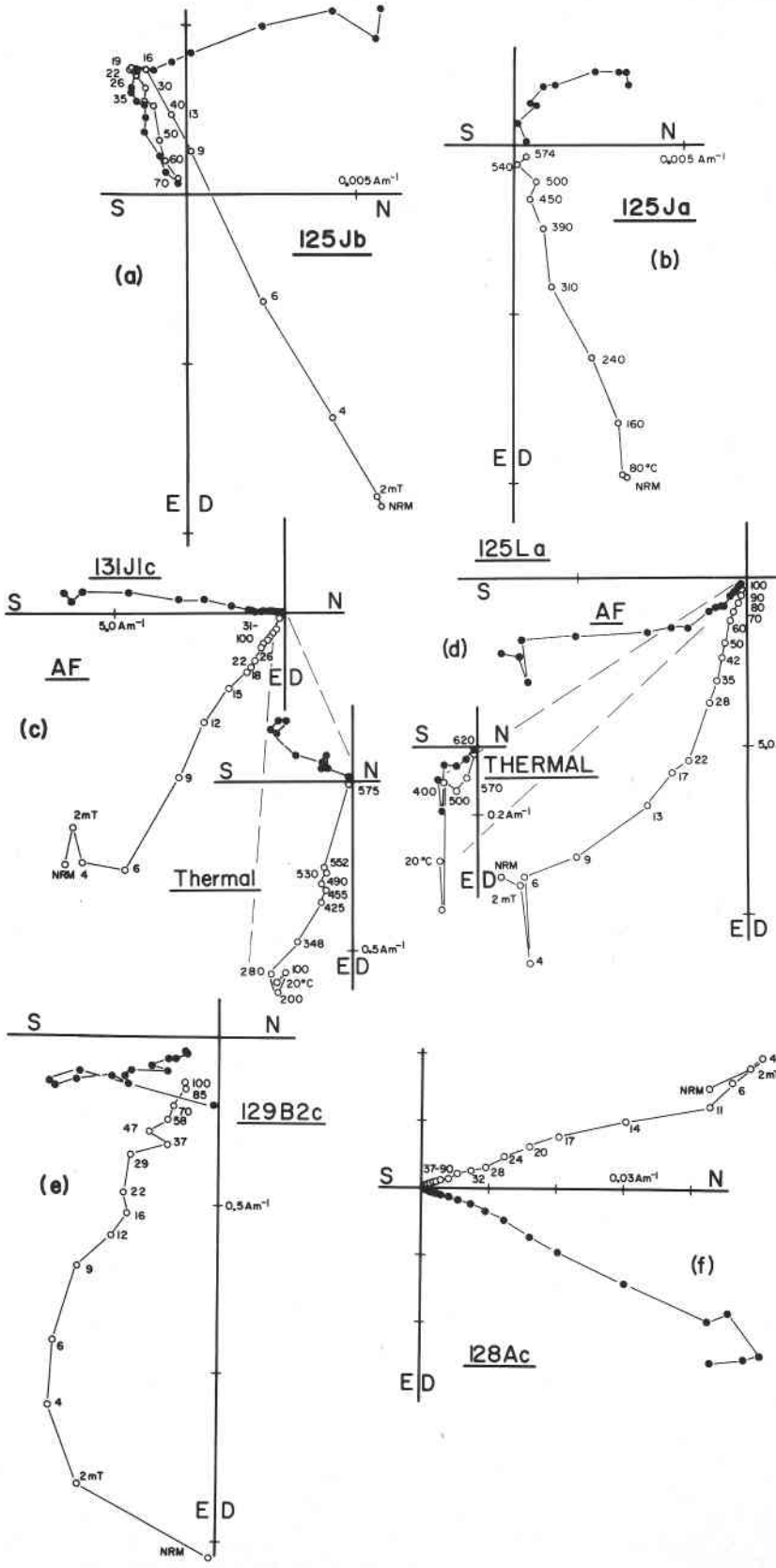
Laboratory storage tests (periods of greater than 1 month, with the NRM direction perpendicular to the Earth's field) indicate that the secondary components common to the low-iron flow are not viscous in origin. The secondary components are removed by approximately 15.0 mT or 250°C in demagnetization. Unfortunately, these components themselves are not well grouped and therefore cannot be correlated with younger thermotectonic events.

Thackeray Township

In Thackeray Township (Fig. 1), no Matachewan dikes are exposed or inferred from aeromagnetic data. The area consists of a thick succession of high-iron flows overlain to the south by a succession of low-iron flows. Two sampling traverses were made through the exact flows described by Pearce and Birckett (1974). One was through a single, complete, high-iron flow in which 38 samples were collected (sites 125 and 131, Fig. 6). At the other traverse, through a portion of a low-iron flow, 23 samples were collected (sites 128 and 129, Fig. 6). In addition, in the central part of this area (Fig. 6), compaction features and contacts between high-iron tholeiite flows suggest that one block has been rotated nearly 80°. Six samples were collected from this area (site 130) with the hope of recognizing this rotation.

For the high-iron flow, the NRM was again found to be often multivectorial (Fig. 7 *a* and *b*). In many samples, where two or more components of magnetization were resolved in AF demagnetization, thermal demagnetization again yielded only one, indicating the substantial overlap of blocking-temperature spectra and implying that the secondary components (fully isolated in alternating field demagnetization) may be chemical in origin. The AF demagnetization was thus the preferred method of resolving all components. In AF treatment, however, the component trending toward the origin was not always completely removed at 100 mT. To ascertain if magnetite rather than secondary hematite is the carrier of the remanence, several samples were demagnetized by AF method to between 60 and 100 mT, then thermally demagnetized in vacuum or air beginning at room temperature. Two examples of the results are shown in Fig. 7 *c* and *d*. Sample 131J shows a nearly linear decay

FIG. 7. Progressive orthogonal demagnetization diagrams (Zijderveld 1967). Conventions are the same as those in Fig. 3. Demagnetizations of (*a*, *b*) two specimens from the same sample in the high-iron flow at site 125; (*c*) high-iron flow specimen from sample J at site 131 (AF followed by thermal demagnetization); (*d*) high-iron flow specimen from sample L at site 125 (AF followed by thermal demagnetization); (*e*) low-iron specimen from sample B at site 129; and (*f*) low-iron basal tholeiite specimen from sample A at site 128.


125Jb
(a)
S
N
0.005 Am⁻¹
E D
2mT
NRM
125Ja
(b)
574
540
500
450
390
310
240
160
80°C
131J1c
(c)
AF
5.0 Am⁻¹
Thermal
575
552
530
490
455
425
348
280
200
0.5 Am⁻¹
125La
(d)
THERMAL
620
570
400
500
0.2 Am⁻¹
20°C
129B2c
(e)
128Ac
(f)
0.03 Am⁻¹

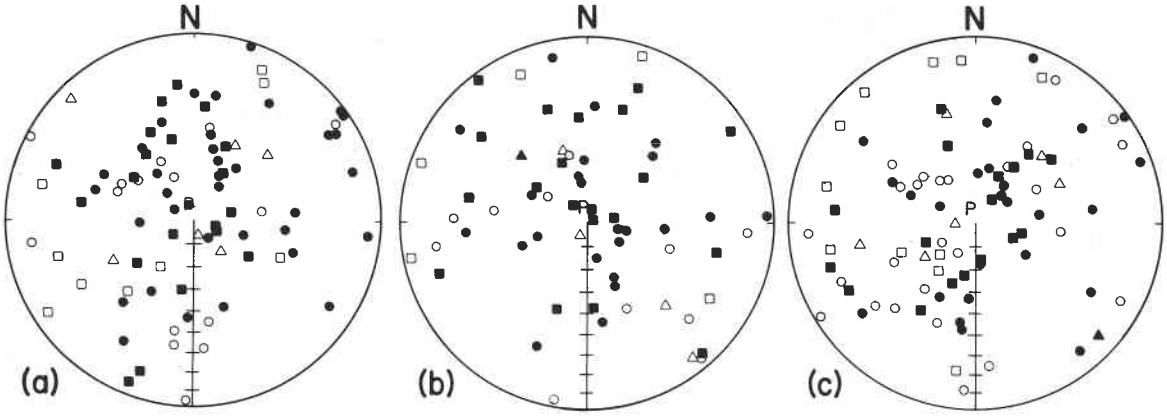


FIG. 8. Paleomagnetic data from sites in Thackeray Township. Conventions are as in Fig. 5. (*a*) NRM directions. (*b*) Any secondary directions. (*c*) Ultimate directions. Circles refer to directions from Mg tholeiite flow, squares from the Fe tholeiite flow, and triangles from site 130.

of a magnetization component in both treatments; essentially no remanence remains by 575°C. Sample 125La possesses a southerly declination, shallow-inclination component (removed by 22 mT) and a southeast-declination, steeper inclination component removed in AF then thermal demagnetizations. Some of the remanence remains above maximum magnetite blocking temperatures, but it is identical to that carried in magnetite. Coercivity of remanence (H_r) values for these samples are less than 20.0 mT (Table 1) and we conclude that the vast majority of the remanence indeed resides in magnetite.

Magnetization data from the high-iron flow are shown in Fig. 8 *a–c*. The directions of the components trending toward the origin in high peak fields or temperatures (Fig. 8*c*) are random at a 95% confidence level (Fisher 1953). For lack of a universal terminology, we will refer to such components as *ultimate* components (discussion in Tasillo-Hirt *et al.* 1982). They are highly "stable" magnetically, but the scatter suggests that they do not represent any one period of geologic time.

For the low-iron tholeiite flow sampled, a similar pattern of magnetization directions was observed. In general, high-coercivity, high-blocking-temperature ultimate components are present in all samples (Fig. 7 *e* and *f*). Neither the ultimate nor the secondary components from the low-iron flow appear to have recorded a unique geologic event (Fig. 8 *b* and *c*).

The data at site 130, in the high-iron flows (Fig. 8 *a–c*), are as dispersed as those from the two flow sequences described above. The results could not be used to test for the rotation of the block.

McVittie Township

A portion of a low-iron Kinojevis flow was sampled (42 samples) in McVittie Township, on the south limb of the synclinorium (Fig. 1). As in Thackeray Township, no Matachewan dikes are exposed or inferred on the basis of aeromagnetic data (Thompson 1941). The NRM varies throughout the flow from a single component of magnetization resolved in either AF or thermal demagnetization (Fig. 9*a*) to a complex, multicomponent magnetization (Fig. 9 *b* and *c*). Thermal demagnetization following AF treatment verified that the remanence resides in magnetite (Fig. 9*d*). In samples containing abundant pyrite, oxidation of the sulfide during demagnetization at temperatures in excess of 300°C, even under vacuum, created significant magnetic material (maghemite, then hematite) and precluded the complete demagnetization of the NRM over magnetite blocking temperatures.

The magnetic directions from this traverse are shown in Fig. 10 *a–c*. The NRM and ultimate directions, in comparison with results from Thackeray Township, are better grouped, with southwest declinations prevalent. Although the NRM directions are less dispersed than the ultimate directions ($k = 2.6$ for NRM versus $k = 1.9$ for ultimate directions), storage tests for periods of greater than 1 month (with the NRM direction perpendicular to the Earth's field), the general presence of multicomponent magnetizations, and the grouping of NRM directions away from a present field value all suggest that the remanence at this site is not dominated by viscous components.

High-coercivity, high-blocking-temperature magnetization components are present in the majority of samples analyzed. In most cases, however, the directions of these components are scattered, and do not reflect a short period of time, as would be expected by the TRM of an individual flow.

Magnetic mineralogy

We contend that the primary (magmatic) magnetic mineral in the Kinojevis tholeiites was probably titanomagnetite having a restricted composition range (close to that of fresh modern layer 2A oceanic tholeiites

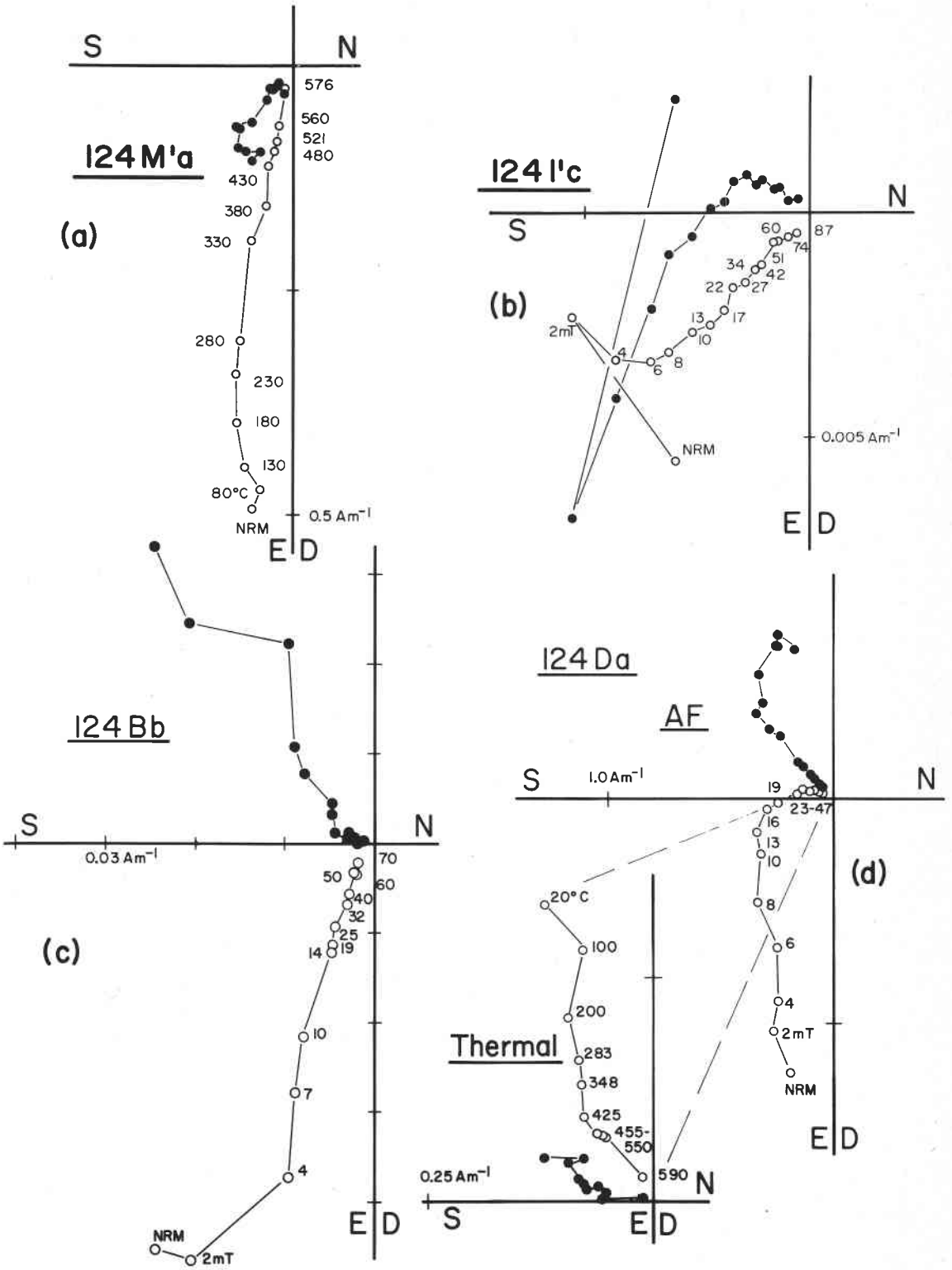


FIG. 9. Progressive orthogonal demagnetization diagrams (Zijderveld 1967). Conventions are the same as those in Fig. 3. Demagnetizations of specimens from samples of the low-iron flow in McVittie Township (Fig. 1).

TABLE 1. Typical rock magnetic and remanence properties, Kinojevis Group tholeiites

Sample	Rock type	H_c (mT)	H_{CR} (mT)	J_s* (A m^{-1})	J_r/J_s*	T_C (°C)	X_i	X_p	NRM (A m^{-1})	$H_{1/2}$ (mT)	T_B (°C)
26A	High-Fe tholeiite, flow base, Cook	—	—	—	—	—	0.86×10^{-5}	—	0.035	50.0	380–540
27F	High-Fe tholeiite, flow center, Cook	3.5	—	65 900.0	0.04	568	4.05×10^{-3}	3.83×10^{-7}	0.86	5.5	—
28A	High-Fe tholeiite, flow base, Cook	7.0	29.0	61.6	0.15	560	1.11×10^{-4}	2.10×10^{-6}	0.096	14.0	—
29A	High-Mg tholeiite, glomeroporphyritic top, Cook	—	—	3.4	—	230, 570	4.29×10^{-5}	1.90×10^{-6}	0.0067	47.0	—
40D	High-Mg tholeiite, flow top, Cook	5.0	—	14.0	0.21	—	3.43×10^{-5}	2.30×10^{-6}	0.0061	48.0	350–540
40J	High-Mg tholeiite, flow center, Cook	—	—	—	—	—	9.43×10^{-5}	—	0.049	60.0	480–580
40T	High-Mg tholeiite, flow center, Cook	11.0	21.0	5 334.0	0.11	560	9.43×10^{-5}	2.50×10^{-6}	8.2	7.0	—
40BB	High-Mg tholeiite, flow base, Cook	—	—	—	—	560	8.57×10^{-5}	—	0.120	9.0	410–580
41B	High-Mg tholeiite, glomeroporphyritic top, Cook	16.0	160.0	126.0	0.02	—	2.20×10^{-5}	3.00×10^{-6}	0.0098	36.0	300–580
124B	High-Mg tholeiite, glomeroporphyritic top, McVittie	3.7	—	5 180.0	0.02	583	1.21×10^{-3}	6.90×10^{-7}	0.52	8.0	NRM-400
124C	High-Mg tholeiite, flow top, McVittie	8.1	—	80 780.0	0.06	575	1.44×10^{-3}	6.67×10^{-6}	10.30	7.5	NRM-580
124E	High-Mg tholeiite, flow top, McVittie	7.0	—	14 420.0	0.08	572	6.98×10^{-3}	1.83×10^{-6}	11.40	7.6	—
124K	High-Mg tholeiite, flow center, McVittie	6.1	—	2 576.0	0.06	576	4.43×10^{-3}	9.00×10^{-7}	30.1	4.0	NRM-580
124BB	High-Mg tholeiite, flow base, McVittie	2.6	8.0	14 600.0	0.03	578	1.19×10^{-2}	6.71×10^{-6}	60.3	11.0	—
124II	High-Mg tholeiite, flow base, McVittie	8.0	100.0	117.6	0.04	—	5.14×10^{-5}	8.30×10^{-7}	0.0055	15.0	NRM-550
125A	High-Fe tholeiite, flow center, Thackeray	—	0	14.1	0.06	578	5.14×10^{-5}	8.10×10^{-6}	0.031	18.0	NRM-580
125B	High-Fe tholeiite, flow center, Thackeray	4.5	25.0	742.0	0.08	577	3.00×10^{-4}	2.00×10^{-6}	0.291	7.0	—
125K	High-Fe tholeiite, flow center, Thackeray	9.5	—	182.0	0.26	555	8.57×10^{-5}	1.60×10^{-6}	0.40	14.5	280–590
125M	High-Fe tholeiite, flow center, Thackeray	1.0	—	8 208.0	0.10	575	9.43×10^{-5}	3.50×10^{-6}	9.80	5.5	300–580

TABLE 1. (*Concluded*)

Sample	Rock type	H_c (mT)	H_{CR} (mT)	J_s* (A m^{-1})	J_r/J_s*	T_C (°C)	X_i	X_p	NRM (A m^{-1})	$H_{1/2}$ (mT)	T_B (°C)
125N	High-Fe tholeiite, flow center, Thackeray	7.3	—	10 659.0	0.11	575	1.41×10^{-3}	2.50×10^{-6}	12.1	6.5	180–580
125O	High-Fe tholeiite, flow center, Thackeray	4.1	—	342.0	0.17	578	2.49×10^{-4}	4.71×10^{-6}	4.2	7.0	—
125P	High-Fe tholeiite, flow center, Thackeray	6.6	78.0	98.0	0.064	569	2.50×10^{-4}	6.31×10^{-6}	3.20	7.0	200–580
125T	High-Fe tholeiite, hyaloclastite base, Thackeray	6.1	48.0	28.0	0.24	580	1.89×10^{-4}	4.51×10^{-6}	0.43	13.0	180–580
131F	High-Fe tholeiite, flow top, Thackeray	2.1	—	168.0	0.06	576	—	6.00×10^{-6}	0.0060	30.0	450–580
131I	High-Fe tholeiite, flow top, Thackeray	5.5	—	8.8	0.13	560	—	1.61×10^{-6}	0.21	12.0	—
131K	High-Fe tholeiite, flow center, Thackeray	16.8	44.0	2 058.0	0.20	574	—	9.01×10^{-6}	4.10	11.0	200–575
128B	High-Mg tholeiite, flow base, Thackeray	6.1	—	203.0	0.08	573	2.57×10^{-4}	2.31×10^{-6}	0.065	4.0	200–580
129B	High-Mg tholeiite, flow center, Thackeray	3.8	0	6 160.0	0.03	575	1.11×10^{-2}	5.53×10^{-6}	3.81	8.0	400–580
129F	High-Mg tholeiite, flow top, Thackeray	3.0	18.0	490.0	0.09	—	4.63×10^{-4}	2.41×10^{-6}	0.376	15.0	300–480
300B	Matachewan dike, Cook	—	—	—	—	—	—	—	0.223	19.0	350–580
300C	High-Fe tholeiite, adjacent to dike, Cook	—	—	—	—	—	—	—	0.087	63.0	400–580
300D	High-Fe tholeiite, adjacent to dike, Cook	—	—	—	—	—	—	—	7.6	12.0	—

NOTES: H_C = coercivity; H_{CR} = coercivity of remanence; J_s = saturation magnetization; J_r/J_s = ratio of saturation remanence to saturation magnetization; T_C = Curie temperature; X_i = initial susceptibility; X_p = paramagnetic susceptibility; NRM = intensity of the natural remanent magnetization; $H_{1/2}$ = median destructive field; T_B = blocking temperature range of the remanence decaying to the origin in thermal demagnetization.

*J_s and J_r/J_s values were often difficult to determine accurately because of the high (10^{-6} or greater) values of paramagnetic susceptibility in conjunction with low saturation magnetization values for many samples.

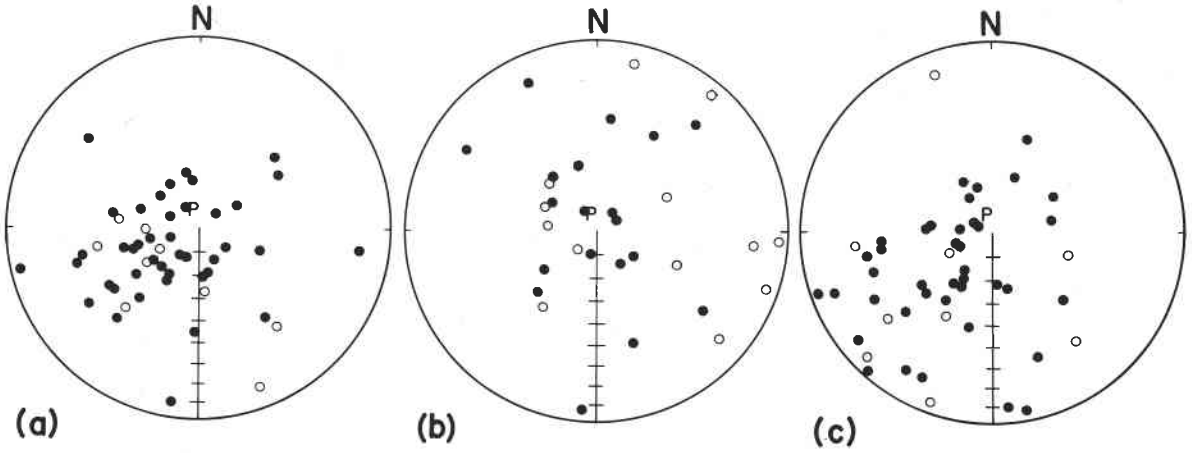


FIG. 10. Paleomagnetic data from site 124 in McVittie Township. Conventions are as in Fig. 5. (*a*) NRM directions. (*b*) Any secondary directions. (*c*) Ultimate directions.

(where $x = 0.6–0.65$ in $Fe_{3-x}Ti_xO_4$), Johnson and Hall 1978). Steiner (1982), however, has demonstrated a significant variability in ulvospinel contents (x = 0.5–0.75) in nearly fresh titanomagnetites of ocean floor basalts and diorites. The originally high titanium content of Kinojevis titanomagnetites has been confirmed in the present study (to be discussed shortly) and by Hanes (1973) who reported ulvospinel contents as high as 63% in titanomagnetites from Kinojevis flows. The Kinojevis basalts would then have had Curie temperatures below 200°C. Unlike modern layer 2A tholeiites, individual Kinojevis flows are thicker, more extensive, and more variable in chemistry than ocean floor tholeiites. The Kinojevis units thus have undergone chemical and mineralogical differentiation *in situ*, as well as important deuteric changes and interactions with seawater like modern ocean floor basalts. Cumulate silicate textures are indeed common in the interiors of flows (Pearce and Birckett 1974). The primary magnetic mineralogy and subsequent alteration products vary significantly within the Kinojevis units. *In situ* differentiation and deuteric (local) alteration appear to be the causes of these variations, but regional burial is probably the most influential cause of the overall difference in magnetic properties between the Kinojevis tholeiites and layer 2A rocks.

Results of alternating field and thermal demagnetization studies indicate that magnetite is the essential magnetic phase. Rock magnetic data are summarized in Table 1. Representative (at least triplicate) samples of all units were analyzed thermomagnetically and in most cases, excluding those where paramagnetic material completely dominated the J_s–T curve, the presence of magnetite was confirmed (Fig. 11). Curie temperatures of the main magnetite phase were always between 560 and 585°C. In very rare instances (e.g., the glomeroporphyritic plagioclase-rich layer in the low-iron flow samples at Cook Township), a ferromagnetic phase of lower Curie temperature (higher Ti content?) was also indicated (Fig. 11, sample 29A).

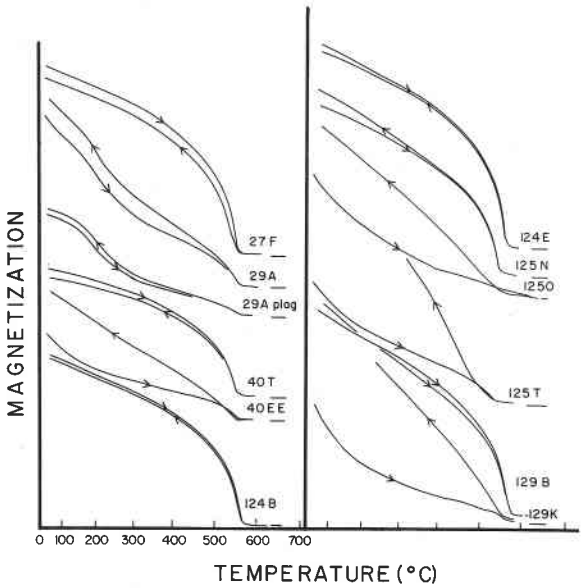


FIG. 11. Saturation magnetization versus temperature (thermomagnetic) curves for representative Kinojevis whole-rock samples. Additional Curie-temperature data are given in Table 1. The saturating field is usually 0.2 T; the vacuum was approximately 10^{-3} Torr (0.13 Pa).

Progressive IRM acquisition and subsequent backfield direct current (DC) demagnetization (Park 1975) of IRM for representative samples also showed magnetite to be the essential magnetic phase, and relatively little hematite was detected at higher fields in saturation (Table 1).

The abundance and textural relations of the microscopically visible magnetic mineralogy of the Kinojevis tholeiites vary considerably, both within and between flow units. The best preserved textural relations occur in basal hyaloclastite and upper, fragmental portions of flow units. For example, in the low-iron flow at Thackeray Township (sample 129B, Fig. 12 *a* and *b*) completely fresh, euhedral, lath-shaped ilmenite and equidimensional magnetite grains are aligned with groundmass material and are compressed and deformed around shards formed by the distintegration of early, quenched glassy material. Representative magnetite compositions determined by electron microprobe (Table 2, analysis 1) are too low in Ti (in agreement with the Curie-temperature data) to have been directly precipitated as homogeneous phases from a tholeiitic basalt. Homogeneous ilmenite laths (Table 2, analyses 2 and 3) are high in hematite component, however, suggesting a high temperature of formation (Buddington and Lindsley 1964; Lindsley 1977). Because both intergrowths and individual minerals are aligned parallel to primary compaction and flow features, often display crystallization relations with early formed silicate phases, and are not spatially related to low-grade silicate devitrification–alteration products, we infer that the oxide phases were formed contemporaneously with the flows.

Magnetite and ilmenite are also quite fresh, albeit rare, in glomeroporphyritic layers common to low-iron Kinojevis flows. For example, in the McVittie Township flow sampled, fine, euhedral, relatively Ti-poor magnetite grains (Table 2, analysis 4) occur in the groundmass (Fig. 12*c*).

In other parts of flow units, sizes and textures of magnetite grains are more varied. In the coarse-grained high-iron flow in Thackeray Township, magnetite grains display intercumulus outlines (Fig. 12*d*) or quenched cruciform (Haggerty 1976, p. HG-106) textures (Fig. 12*e*), or are found as inclusions in silicate phases (Fig. 12*f*). Sporadic portions of the low-iron McVittie flow and the high-iron Cook flow are extremely enriched in subhedral magnetite grains of diverse sizes (Fig. 12*g*). The distribution of magnetite grains in these enriched areas, conformable with flow textures in the groundmass, suggests that they were precipitated simultaneously with the rest of the surrounding tholeiite melt in an irregular fashion.

The amount of ferromagnetic material in a flow is a function of both the primary crystallization–differentiation history and subsequent oxidation. Uniform variations of magnetization parameters (e.g., NRM intensity, intensity behaviour in alternating field demagnetization, Fisher precision parameter, and oxidation index) would be expected within a flow unit were both processes regular (Watkins and Haggerty 1967*a*,*b*). A measure of the relative amount of ferromagnetic material in a rock and its variability is the ratio X_p/J_s (Dunlop and Hale 1977) where X_p is the paramagnetic susceptibility (due largely to ferromagnesian silicates) and J_s is the saturation magnetization. The paramagnetic susceptibility is measured between 300 and 800 mT in hysteresis (Collinson 1968) and J_s is the y intercept of the tangent to the magnetization–H curve at maximum H. Dunlop and Hale (1977) argued that if paramagnetic and ferromagnetic materials are comagmatic, i.e., deriving iron from the same sources, then the ratio X_p/J_s should be relatively constant. Figure 13 shows the results of this test for the traverse through the complete section of the Thackeray high-iron flow. The X_p/J_s ratios do not show a good correlation with lithologic variations. The variability suggests that alteration events have irregularly affected the magnetic mineralogy.

On the other hand, X_p/J_s ratios from the low-iron McVittie flow are all very low, ranging from 0.44×10^{-5} to 4.8×10^{-5}. Magmatic and (or) alteration processes affecting the magnetic mineralogy must have been relatively uniform through the portion of the flow sampled.

In general, alteration of the primary magnetic mineralogy appears to increase toward the centers or lower centers of flows, together with an increase in silicate grain size. In the central part of the low-iron flow at Cook Township and the high-iron flow at Thackeray Township magnetite is abundant as a coarse, intercumulus phase but has invariably undergone oxidation to hematite, rutile, and sphene (Fig. 12*d*). Relict structures indicative of ilmenite–magnetite intergrowths probably formed during earlier, high-temperature deuteric oxidation–exsolution. The degree of clearly secondary, low-temperature oxidation varies even on a thin-section scale, and irregular patches of titanomagnetite (Table 2, analysis 5) commonly occupy centers of original titaniferous magnetite grains (Fig. 12*d*). Small magnetite inclusions in silicate phases appear homogeneous and unoxidized under the microscope.

The most important aspect of the magnetic mineralogy of the Kinojevis basalts is the ubiquitous property of Curie temperatures greater than 550°C, indicative of nearly pure magnetite. Electron microprobe data do not always support this observation, however. For example, most magnetite grains in the low-iron McVittie flow are homogeneous at the resolution of the optical microscope (approximately 1 μm) and display shrinkage cracks at their edges, suggesting partial maghemitization. Some grains clearly display oxidation–exsolution intergrowths (Fig. 12*h*). The optically homogeneous grains are both chemically homogeneous at the electron microprobe level *and*

TABLE 2. Magnetite and ilmenite compositions from selected Kinojevis units

	Sample 129B			Sample 124A		Sample 125N	Sample 124E		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
TiO_2	0.27	49.50	47.83	0.35	6.63	24.58	28.41	26.06	0.66
Al_2O_3	0.32	0.00	0.00	0.17	0.27	0.00	0.16	0.00	0.39
FeO*	93.28	42.61	42.96	89.45	84.70	70.85	67.20	69.58	89.23
FeO	25.21	37.43	37.88	31.31	39.02	54.31	57.27	54.65	35.54
Fe_2O_3†	68.06	5.72	5.58	63.92	50.21	18.16	10.91	16.41	59.03
MnO	0.00	6.34	5.65	0.00	0.58	0.00	0.00	0.00	0.00
MgO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
V_2O_5	—	—	—	0.00	—	0.66	0.77	0.70	0.25
SiO_2	—	0.40	0.62	0.86	0.81	0.60	0.77	0.54	2.45
Sum‡	100.03	99.40	97.56	96.61	97.52	98.32	98.29	98.36	98.32
Formulae normalized about three (spinel) or two (rhombohedral) cations									
Fe^{2+}	1.01	0.80	0.83	1.04	1.27	1.73	1.81	1.74	1.15
Fe^{3+}	1.97	0.11	0.11	1.91	1.47	0.52	0.31	0.47	1.72
Ti	0.01	0.96	0.94	0.01	0.19	0.70	0.81	0.75	0.02
Mn	0.00	0.14	0.13	0.00	0.02	0.00	0.00	0.00	0.00
Mg	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Al	0.01	0.00	0.00	0.01	0.01	0.00	0.01	0.00	0.02
V	—	—	—	0.00	—	0.02	0.02	0.02	0.01
Si	—	0.01	0.02	0.03	0.03	0.02	0.03	0.02	0.10
Mol% ulvospinel (spinel) or hematite (rhombohedral)									
	1.01	6.3	6.4	1.0	19.7	72.9	83.9	76.1	2.3

*Total FeO as determined by electron microprobe.

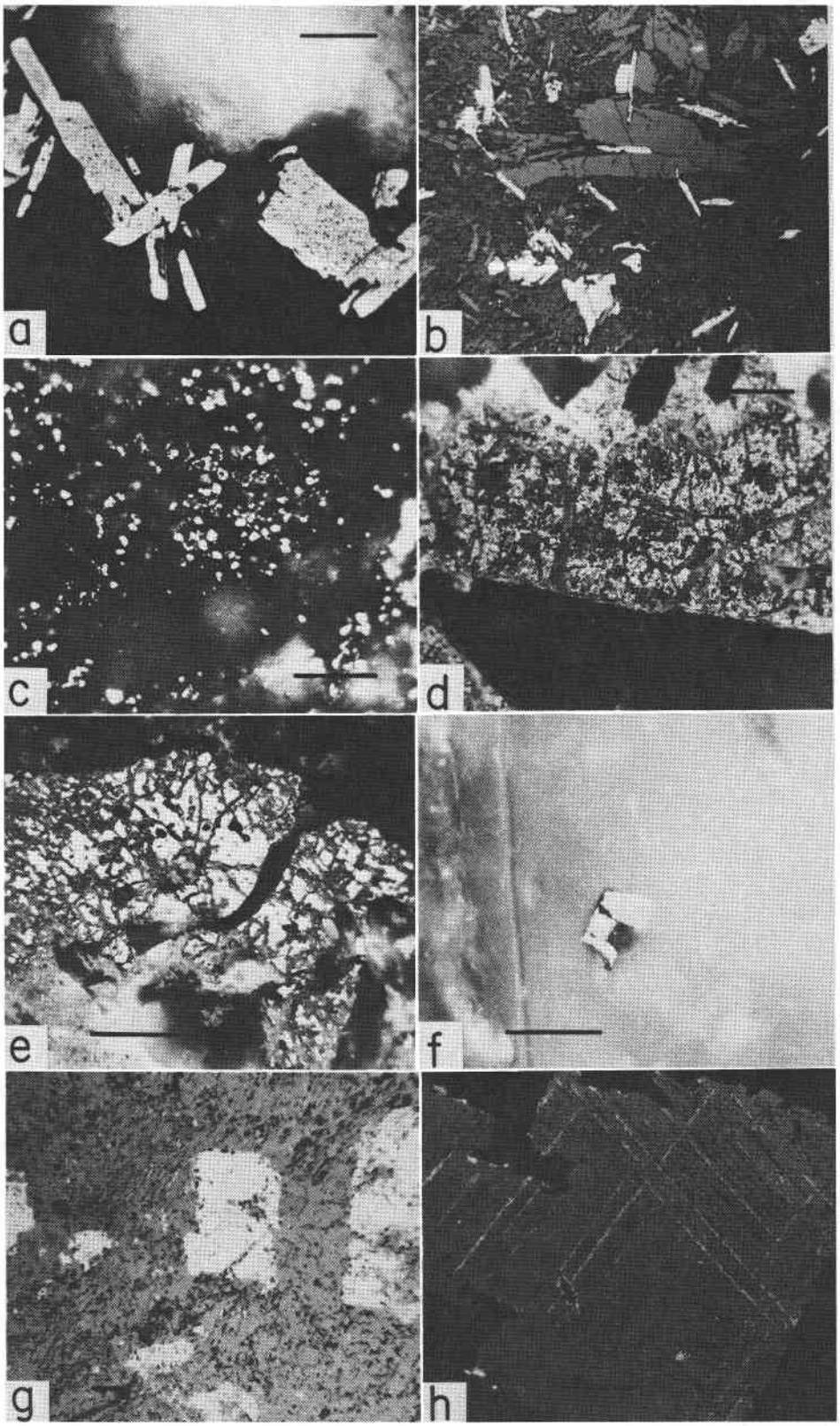
†The amount of Fe_2O_3 is calculated from stoichiometry.

‡The sum of the oxides is calculated using the calculated weight per cent Fe_2O_3. The method of calculating end-member percentages along the spinel and rhombohedral solid solution joins from the number of moles of an element per formula unit is essentially that of Bohlen and Essene (1977).

titanium rich (Table 2, analyses 6–8). Edges of grains are often poorer in Ti (Table 2, analysis 9), suggestive of Ti mobilization during maghemitization. There is no indication, however, of a low-Curie-temperature (high-Ti) phase in any sample from this flow (Table 1, Fig. 11). Powder X-ray diffraction work on magnetic separates from several Kinojevis samples also indicates the abundance of a titanium-poor, non-cation-deficient spinel phase. The grains have possibly undergone phase splitting (to ilmenite and nearly pure magnetite) on a submicron scale, a phenomenon rarely observed in ocean floor basalts (Rice *et al.* 1980). The inversion and the production of a CRM in secondary magnetite could have occurred over a prolonged period of time, as indicated by results of experimental studies on titanomaghemite (Johnson and Hall 1978; Ryall and Hall 1979; Keefer and Shive 1980). Magnetic grains analyzed by Hanes (1973) give a range of ulvospinel contents from 63 to 10 mol.%, also lending credence to the possibility of extensive cation migration.

Oxidation–exsolution of titanomagnetites at high temperatures may have been precluded by relatively

FIG. 12. Photomicrographs of representative magnetic oxide parageneses in the Kinojevis flows. In each photomicrograph the scale bar represents 20 μm. (*a*) Euhedral ilmenite laths in hyaloclastite, sample 129K, high-Mg tholeiite, Thackeray Township. (*b*) Euhedral ilmenite laths and subhedral to euhedral magnetite grains in hyaloclastite, sample 129B, high-Mg tholeiite, Thackeray Township (the large amphibole). (*c*) Euhedral to subhedral magnetite grains in the groundmass of the upper, glomeroporphyritic portion of the high-Mg flow sampled in McVittie Township, sample 124C. (*d*) Highly oxidized magnetite grain displaying interstitial and poikilitic textural relations with silicate grains, center of high-Fe tholeiite flow sampled in Thackeray Township, sample 125H. (*e*) Interstitial partly oxidized magnetite grain, center of high-Fe tholeiite, Thackeray Township. (*f*) Fresh subhedral magnetite inclusion in plagioclase, sample 125H, high-Fe tholeiite, Thackeray Township. (*g*) Euhedral (optically homogeneous) magnetite grains in the high-Mg tholeiite flow sampled in McVittie Township, sample 124E. (*h*) Magnetite–ilmenite oxidation–exsolution intergrowth in sample 124N, high-Mg tholeiite, McVittie Township.


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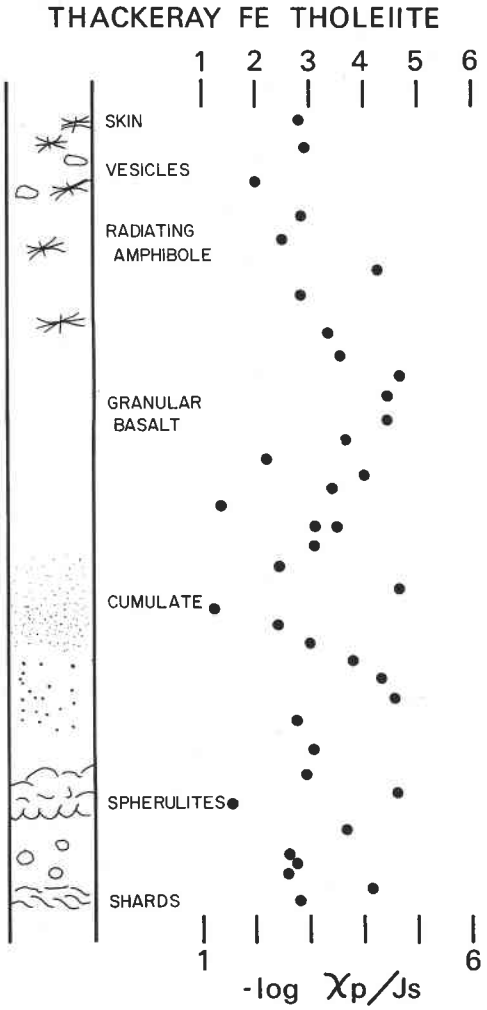


FIG. 13. Values of X_p/J_s, (paramagnetic susceptibility)/(saturation magnetization), plotted against stratigraphy, high-iron tholeiite flow (sites 125, 131) in northern Thackeray Township.

reducing conditions imposed by high $f S_2$ values and the eventual precipitation of pyrite below 730°C. Titanomagnetites in regions devoid of pyrite or containing pyrite deposited in late-stage, low-temperature conditions show evidence of more intense high- and low-temperature oxidation.

The proposed action of phase splitting clearly distinguishes the Kinojevis flows from typical modern ocean floor tholeiites. *In situ* ocean floor lavas with high Curie temperatures have been rarely reported (e.g., Ryall *et al.* 1977; Rice *et al.* 1980). In contrast, work on ophiolite complexes has shown an abundance of high-Curie-temperature basalts (Butler *et al.* 1975; Levi and Banerjee 1977; Levi *et al.* 1978; Beske-Diehl and Banerjee 1979; Banerjee 1980). These results have been interpreted as indicating a reheating (metamorphic) event causing the unmixing of titanomaghemites at some later, but unknown, time (Levi *et al.* 1978). High-temperature oxidation probably occurred in some Kinojevis flows because the flows were too massive and deposited and buried too quickly to be quenched instantaneously and crystallize fine-grained titanomagnetite.

The magnetic mineralogy of the Kinojevis basalts has been more extensively altered (probably involving pervasive phase splitting of homogeneous titanomaghemites, produced from original titanomagnetites) than that of normal ocean floor tholeiites; the remanence is entirely secondary in origin.

Discussion

Interpretations of the paleomagnetic data from the Kinojevis tholeiites must involve three important features. First, individual samples usually contain at least one magnetization component of intermediate to high blocking temperatures and coercivities. These components generally display linear decays in demagnetization and therefore represent a relatively discrete period of geologic time. Second, "contact" tests (this study; Pesonen 1973) indicate the ability of at least some units to retain a statistically well grouped secondary TRM imparted by Matachewan dikes. Additional magnetization components, some perhaps chemical in origin, have been superimposed upon the overprinted TRM. Third, we suggest that original titanomagnetites of all flows have been in part converted to nearly pure magnetites in a probably complex series of changes.

As noted previously, the thermal history of this area is complex, but critical information bearing on the magnetization features listed above is available. Maximum regional metamorphic temperatures could have been less than 400°C; the upper stability limit of prehnite–chlorite–quartz assemblages (very common in Kinojevis units) is approximately 300°C at a pressure of 3 kbar (300 MPa), and decreases at higher pressures (Nitsch 1971). Schwarz and Buchan (1980), Buchan and Schwarz (1981), and L. D. Schutts (personal communication, 1980) reported that the ambient temperature of presently exposed country rock at Munro Township during "Abitibi" dike intrusion (approximately 2150 Ma) was between 181 ± 7 and 250 ± 25°C, implying a prolonged deep burial for this area. Schwarz and Buchan (1980) and Buchan and Schwarz (1981) also suggested that the Abitibi Belt remained deeply buried through Middle Proterozoic time. Magnetic (Schutts and Dunlop 1981) and ^{40}Ar–^{39}Ar (Hanes 1979) data further suggest mild thermal activity in the Abitibi area during Hudsonian (approximately 1800–1600 Ma) orogenic activity. Assuming a maximum duration of heating of 1.0 Ga, an original remanence residing in

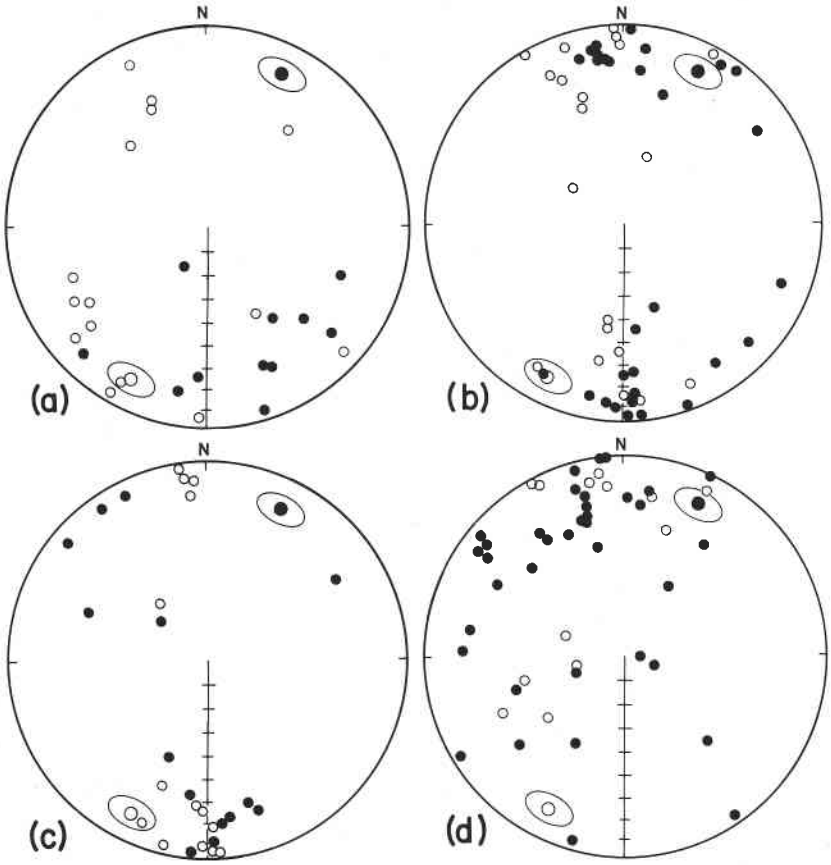


FIG. 14. Ultimate directions of magnetization from Figs. 5*c*, 8*c*, and 10*c* corrected for regional tilt of the Kinojevis flows. Conventions are as those in Fig. 5. In each projection, the normal and reversed polarity Matachewan dike directions are shown with their associated cones of 95% confidence (Irving and Naldrett 1977). (*a*) High-iron flow at Lava Flow Mountain (the Matachewan dike components in the Mg tholeiite are not included). (*b*) High-iron flow in Thackeray Township. (*c*) Low-iron flow in Thackeray Township. (*d*) Low-iron flow in McVittie Township.

essentially pure single-domain magnetite capable of surviving heating must *theoretically* have blocking temperatures in excess of 500°C (Pullaiah *et al.* 1975). For example, the Matachewan dike component present in Kinojevis tholeiites at sites 40 and 41 is isolated in thermal demagnetization between approximately 350 and 580°C. If the Matachewan remanence was a complete TRM due to reheating, then Pullaiah *et al.*'s (1975) curves, if applicable, yield a maximum ambient temperature of 150°C for partial unblocking to 350°C.

If low-titanium magnetite formed during prolonged heating, as might be the case, then the remanence would be far more complex and Pullaiah *et al.*'s (1975) blocking curves may not be applicable. A CRM of high blocking temperatures (greater than 400°C, as a compromise) would reflect the ambient field during its formation (Merrill 1975). Magnetite grains of lower blocking temperatures would remain unblocked until uplift and would either acquire a scattered or well grouped remanence, depending upon the rate of cooling and scale over which cooling took place.

In most Kinojevis units examined, both high- and low-blocking-temperature components are scattered and must have been acquired over a significant period of time; the formation of low-titanium magnetite continued during burial and metamorphism. Importantly, we have observed that the scatter of directions is not related to the general ranges of blocking temperatures or coercivities over which the magnetization components are removed. On the basis of electron microprobe results from oxides in Kinojevis units and Matachewan dikes in Pesonen's (1973) study area, Hanes (1973) suggested that most titanium loss (and phase splitting) took place prior to Matachewan dike emplacement. This observation supports Jensen's (1981) belief that downwarping and burial of the major Noranda–Benoit synclinorium occurred concurrently with the emplacement of the latest Kinojevis units and the Blake River Group.

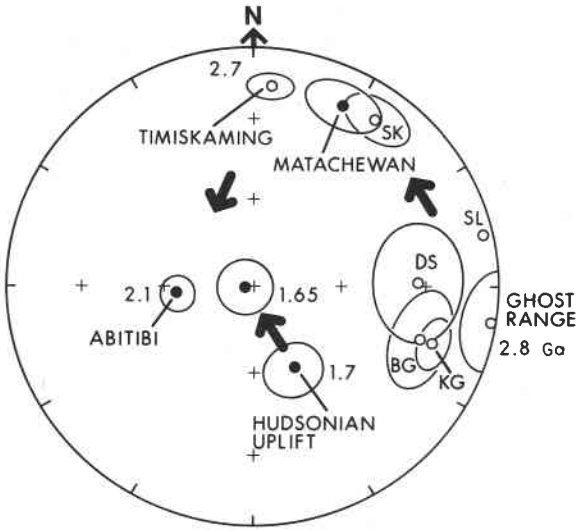


FIG. 15. Field directions (Abitibi coordinates) for Late Archean – Early Proterozoic time, North American craton. KG = Kamiskotia Gabbro (Irving and Naldrett 1977); DS = Dundonald sill (Irving and Naldrett 1977); SL = Shelley Lake Granite, A component (Dunlop and Buchan 1975) (no α_{95} provided); BG = Abitibi basalts and gabbros (Schutts and Dunlop 1981), Timiskaming = Timiskaming non-marine facies sediments (Geissman *et al.*, in preparation); Ghost Range = Ghost Range layered complex (Geissman *et al.* 1982); SK = Skead Group pyroclastics (Ridler and Foster 1981). The Matachewan dike, Abitibi dike, and Hudsonian uplift directions are from compilations by Irving (1979), Irving and Naldrett (1977), and Schutts and Dunlop (1981).

Deformation was essentially complete by the time of Matachewan dike activity.

Tasillo-Hirt *et al.* (1982) and Geissman *et al.* (in preparation) report a characteristic magnetization from Timiskaming sediments near Kirkland Lake that, when corrected for structure, yields a pre-Matachewan dike paleopole. We have likewise corrected the high-blocking-temperature, high-coercivity-magnetization components in samples from Cook, Thackeray, and McVittie Townships for post-emplacement synclinorium deformation (Fig. 14 *a–d*). The data have been corrected assuming simple tilt about a horizontal, nearly east–west striking axis. Additional structural corrections (e.g., to account for an apparent shallow eastward plunge of the synclinorium or possible local structural effects (e.g., Verosub and Moores (1981)) are not used. The eastward plunge of the synclinorium in the sampling area was probably assumed during downwarping of the region, not at a later time, and conventional corrections for this feature may not accurately reflect the true deformation of the units (Jensen 1981). The corrected data in Fig. 14 *a–d* are in only crude agreement with previous observations from Matachewan dikes. Data suggesting a pre-Matachewan dike APW path for the North American craton (Irving and Naldrett 1977; Irving 1979; Schutts and Dunlop 1981; Dunlop and Buchan 1975; Geissman *et al.* 1982) are plotted in terms of field directions for Abitibi coordinates in Fig. 15. Shallow-inclination, but more northeasterly to southeasterly declination directions (as well as antipodes) in Fig. 14 *a–d* may reflect pre-Matachewan positions of the field. We note that our magnetization data from the four sections through Kinojevis flows—*in situ* secondary and *in situ* "ultimate" directions and structurally corrected "ultimate" directions—show little resemblance to the *in situ* magnetization (mean: $D = 114°$, $I = -32°$, $\alpha_{95} = 11$, $k = 15$, $N = 13$ = the number of sites averaged, with no value given for the number of "accepted" samples per site) reported by Schutts and Dunlop (1981). They suggested that their data represent an Archean field existing during deposition and concomitant deformation of basalts and gabbros in the central Abitibi Belt. Our sampling does not overlap with their studies. Because of post-emplacement changes in the magnetic mineralogy, it is conceivable that the Kinojevis flows have sampled a younger Archean field than the units investigated by Schutts and Dunlop (1981). We now explore that possibility.

For this analysis, we assume that (1) proposed phase splitting in the Kinojevis flows took place until (and even possibly later than) Matachewan dike emplacement and (2) the characteristic Matachewan dike direction (approximately $D = 27°$, $I = 16°$, e.g., Irving and Naldrett 1977) accurately represents the paleomagnetic field during changes in the magnetic mineralogy. We compare the structurally corrected Kinojevis ultimate components with theoretical estimates of geomagnetic field behaviour (e.g., Harrison and Watkins 1977; Dodson 1980). The virtual geomagnetic pole (VGP) angular dispersion of the observed directions is significantly greater and more distributed than a theoretical histogram of VGP angular dispersion (Dodson 1980) for the approximate latitude of the Abitibi Belt at latest Archean time (Fig. 16). The Kinojevis units sampled clearly do not yield data consistent with magnetization blocking during distinct, well defined geologic events.

We contend that scatter in directions from individual Kinojevis flows is due to short-term secular variation and (or) polarity reversals in probable combination with long-term polar wander. It is difficult to evaluate the relative contributions of these effects, especially since we observe directional variations within discrete flows and the remanence is not a TRM, or "primary," in the strictest sense. Other investigations of Proterozoic terranes, for example those of Morgan (1976), Beckmann *et al.* (1977), Piper and Stearn (1976), and Piper (1981) in western Greenland, have shown that long-term remanence acquisition records apparent polar wander rather than secular variation, because the cooling–uplift rate is slow enough to average secular

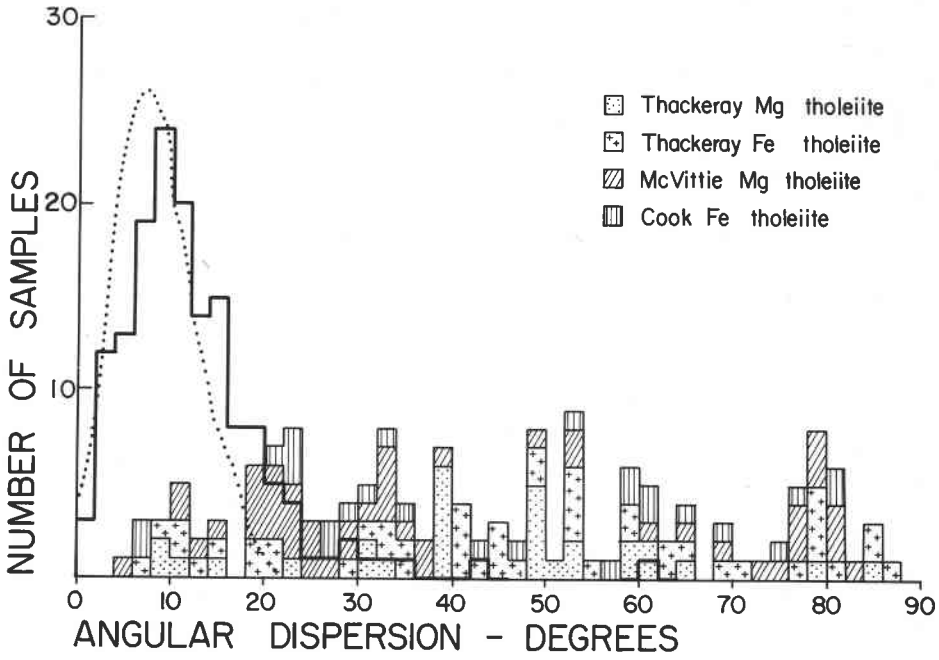


FIG. 16. Histogram of the observed VGP angular dispersion of the ultimate magnetization components ($N = 153$) (corrected for regional structure) in the Kinojevis flows. The heavy solid line is a histogram of theoretical VGP dispersion generated by Dodson's (1980) model 1 at a latitude of 5°N (scaled down for 153 vectors). The dotted curve is the angular dispersion of the geocentric dipole axis.

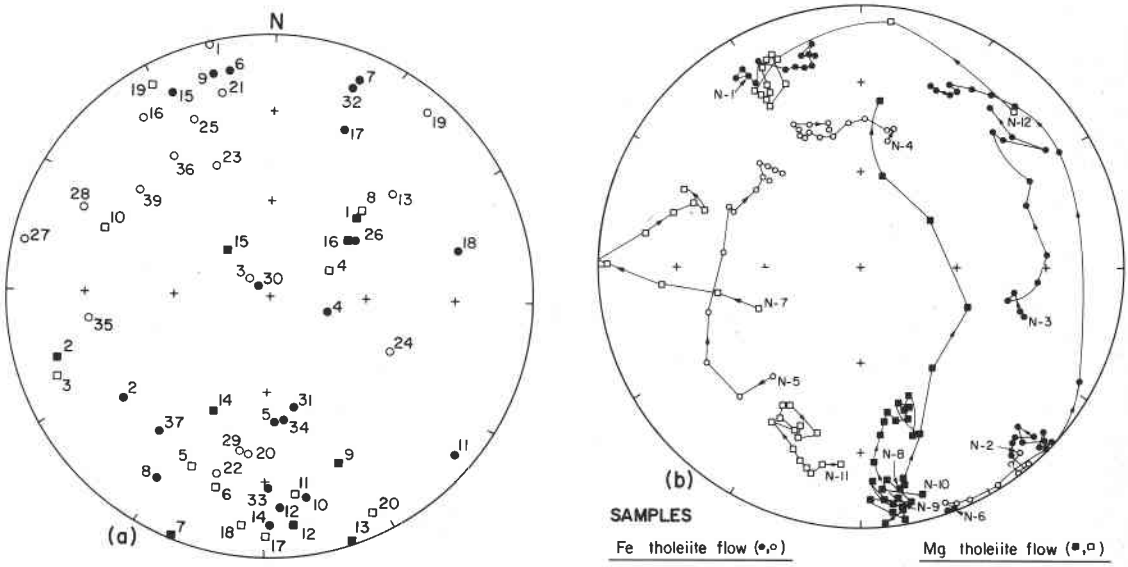


FIG. 17. Paleomagnetic data from the Kinojevis flows at Thackeray Township. (*a*) Directions of ultimate magnetization components, corrected for structure. The directions from samples of each flow are numbered in stratigraphic order, from bottom to top. (*b*) Changes in magnetization directions (corrected for structure) during progressive AF demagnetization of six random samples selected from each flow. Lines join directions of individual samples; arrows refer to increasing peak fields away from the NRM direction.

variation. To investigate the possibility that the results from the Kinojevis units represent long-term polar wander, we have compared (1) the stratigraphic variation in directions of ultimate sample components in the Thackeray Fe tholeiite flow with that in the Thackeray Mg tholeiite flow (Fig. 17*a*) and (2) the overall variation in sample directions with progressive AF demagnetization (Fig. 17*b*). Figure 17*a* illustrates the extremely complex magnetization record in each flow. Few sample directions lie along regular paths and no correlation may be made between directional variations in each flow. On this basis, it appears unlikely that the dispersion of sample directions represents a simple case of systematic polar wander.

Directional variations from individual samples during AF demagnetization are equally complex. Plotted in Fig. 17*b* are the paths of directional changes observed until a linear or nearly linear decay to the origin is obtained. Several samples exhibiting pronounced directional changes give the crude suggestion of blocking during a field change from a reversed polarity direction to a normal polarity one. Field changes may have been sampled by some portions of the Kinojevis flows, but the overall complex nature of the Kinojevis data makes it difficult to assess the role of long-term polar wander in the observed dispersion in the Kinojevis data. It must be remembered that the magnetic mineralogy in the Kinojevis flows is probably completely secondary. Blocking of many, if not most, individual magnetization components as a result may have occurred over geologically short periods of time in the latest Archean.

Conclusions

Basaltic tholeiites of the Kinojevis Group possess an NRM that is generally multivectorial. The components in all units usually reside in pure magnetite, not the originally precipitated magnetic phases. If a TRM was carried by titanomagnetites ($X = 0.65$), it has been lost by chemical reconstitution of the magnetic oxides in response to primary deuteric conditions, long-term burial, progressive regional metamorphism, and possibly intrusion by Matachewan dikes. The directions of the high-coercivity, high-blocking-temperature components, either *in situ* or corrected for downwarping, show poor affinity for any Late Archean to Middle Proterozoic thermotectonic events, even though the corrected directions are crudely dispersed about normal and reversed polarity Matachewan dike directions.

The dispersion of magnetization components must signify that regional blocking took place over a geologically long period of time. Secular variation, field reversals, and long-term polar wander may have contributed to the dispersion. The Kinojevis data do constrain deformation in the central Abitibi Belt to be only slightly older than Matachewan dike intrusion. The Mg tholeiite flow at Cook Township contains a statistically coherent component identical to the TRM of Matachewan dikes. Components of similar coercivity and blocking-temperature spectra in flows elsewhere, when corrected for structure, are in crude agreement with a Matachewan characteristic direction. Extrusion, burial, and tight downwarping of the 10 km or more of Kinojevis stratigraphy all must have occurred under a field very similar to the one recorded by Matachewan dikes. Field and petrologic (Jensen 1981) and other paleomagnetic evidence (Tasillo-Hirt *et al.* 1982; Geissman *et al.*, in preparation) strongly supports this conclusion. Although there appears to be an agreement between the paleomagnetic data and geologic models for greenstone belt evolution in this part of the Abitibi, the Kinojevis data by themselves cannot be taken as statistically reliable indicators of the pre- (or immediately post-) Matachewan geomagnetic field.

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